

# Integrating diverse experimental information to assist protein complex structure prediction by GRASP

Received: 11 October 2024

Accepted: 8 August 2025

Published online: 15 September 2025



Yuhao Xie<sup>1,2,7</sup>, Chengwei Zhang<sup>3,4,7</sup>, Shimian Li<sup>5</sup>, Xinyu Du<sup>5</sup>, Yanjiao Lu<sup>3,4</sup>, Min Wang<sup>6</sup>, Yingdong Hu<sup>6</sup>, Zhenyu Chen<sup>5</sup>, Sirui Liu<sup>1</sup>✉ & Yi Qin Gao<sup>1,3,5</sup>✉

Protein complex structure prediction is crucial for understanding of biological activities and advancing drug development. While various experimental methods can provide structural insights into protein complexes, the knowledge obtained is often sparse or approximate. A general tool is needed to integrate limited experimental information for high-throughput and accurate prediction. Here we introduce GRASP to efficiently and flexibly incorporate diverse forms of experimental information. GRASP outperforms existing tools in handling both simulated and real-world experimental restraints including those from crosslinking, covalent labeling, chemical shift perturbation and deep mutational scanning. For example, GRASP excels at predicting antigen–antibody complex structures, even surpassing AlphaFold3 when using experimental deep mutational scanning or covalent-labeling restraints. Beyond its accuracy and flexibility in restrained structure prediction, GRASP’s ability to integrate multiple forms of restraints enables integrative modeling. We also showcase its potential in modeling protein structural interactome under near-cellular conditions using previously reported large-scale *in situ* crosslinking data for mitochondria.

Protein–protein interactions (PPIs) are essential in multiple biological processes<sup>1</sup>. The disparity between known amino acid sequences and solved protein structures highlights the critical need for computational prediction<sup>2,3</sup>. Current approaches such as AlphaFold-Multimer (AFM)<sup>4</sup>, AlphaFold3 (AF3)<sup>5</sup>, ESMFold<sup>6</sup> and OmegaFold<sup>7</sup> still show limited accuracy in protein complex structure prediction<sup>8</sup>.

Various experimental approaches can provide sparse but valuable information to assist prediction of protein complex structures. Residue pair restraints (RPRs) come from methods such as crosslinking mass spectrometry (XL-MS)<sup>9,10</sup> and nuclear magnetic resonance

nuclear Overhauser effect spectroscopy<sup>11</sup>. Interface restraints (IRs) represent potential interface residues of the complex. Differential covalent labeling (CL) detects reduced covalent modifications when protein complexes transit from unbound to bound states<sup>12</sup>. Chemical shift perturbation (CSP) identifies ligand interacting regions through shifts in chemical signals on binding<sup>13,14</sup>. Deep mutational scanning (DMS) highlights mutations affecting binding, often corresponding to interface residues<sup>15,16</sup>. These methods all provide IR information. In addition, *in situ* biophysical experiments including cryogenic (cryo) electron tomography<sup>17</sup>, solution NMR<sup>18</sup> and *in vivo* XL<sup>19</sup> can detect

<sup>1</sup>Changping Laboratory, Beijing, China. <sup>2</sup>Academy for Advanced Interdisciplinary Studies, Peking University, Beijing, China. <sup>3</sup>Biomedical Pioneering Innovation Center (BIOPIC), Peking University, Beijing, China. <sup>4</sup>School of Life Sciences, Peking University, Beijing, China. <sup>5</sup>New Cornerstone Science Laboratory, Beijing National Laboratory for Molecular Sciences, College of Chemistry and Molecular Engineering, Peking University, Beijing, China. <sup>6</sup>Huawei Technologies Co. Ltd, Hangzhou, China. <sup>7</sup>These authors contributed equally: Yuhao Xie, Chengwei Zhang. ✉e-mail: [liusirui@cpl.ac.cn](mailto:liusirui@cpl.ac.cn); [gaoyq@pku.edu.cn](mailto:gaoyq@pku.edu.cn)

protein interactions at their real functional state. Integrating these experimental data to structure modeling will tremendously benefit the understanding of complicated biological systems.

Several tools were developed to integrate experimental data into protein structure prediction. AlphaLink<sup>20</sup> embeds XL information directly into the pair representation and provides a protein complex version<sup>21</sup>. ColabDock<sup>22</sup> leverages restraints for protein–protein docking by inverting AlphaFold models but is time-consuming and limited to proteins <1,200 residues. Traditional docking methods such as HADDOCK<sup>23,24</sup> and ClusPro<sup>25–27</sup> also support experiment restraints. HADDOCK transforms restraints into energy term to guide conformation sampling and scoring while ClusPro uses them to filter out FFT-generated structures. One study group<sup>28</sup> integrated a CL scoring term into RosettaDock score function to select models aligned with experiments, but is limited to CL data.

In addition, different experimental methods provide cross-validated or complementary information, making them valuable for modeling complicated complexes. Integrative modeling approaches have been developed inspired by this. For example, the integrative modeling platform (IMP)<sup>29</sup> samples protein complex conformations through Monte Carlo simulation and estimates the conditional likelihood of structures given restraints in the Bayesian framework, a framework also advantageous for multi-state modeling<sup>30</sup>. Current integrated modeling approaches are mostly case specific and need much manual effort<sup>31,32</sup>.

In this paper, we introduce a protein complex structure prediction model, the generalized restraints assisted structure predictor (GRASP). GRASP integrates both RPR and IR restraints directly to accommodate diverse experimental data in model inference. It is fine-tuned with restraint-related loss terms and applies iterative noise filtering in inference to reduce the impact of erroneous restraints. We perform comprehensive evaluations on both simulated and real-world experimental data on different functional protein complexes to assess GRASP's ability to handle various types of experimental information, both simultaneously and independently, to improve prediction accuracy. Specifically, we applied GRASP to antibody–antigen complex structure determination, integrative modeling that simultaneously uses multiple experimental data and prediction of *in situ* complex structures in mitochondria. GRASP was shown to provide reliable structures and interactions for complex biological machines, and potentially also molecular details on dynamical protein interactions in cellular environment using experimental information such as XL.

## Results

### GRASP integrates general restraints into protein modeling

GRASP integrates both RPRs and IRs into AFM for protein complex structure prediction. RPRs, derived from experiments such as XL-MS and NMR, provide distance restraints between residue pairs and can be viewed as edge features in a protein complex graph with residues as nodes. IRs, obtained from experimental methods such as CL, CSP and DMS, serve as node features, indicating whether a residue is at the interface. These features are incorporated into AFM via three blocks (Fig. 1a). RPRs are integrated into GRASP as MSA and invariant point attention bias (Fig. 1b) following our previous work<sup>33</sup>, while IRs are introduced using relative position encoding and interface fusion blocks within the Evoformer and invariant point attention modules (Fig. 1c).

We added four restraint-related loss terms (Supplementary Methods) to the original loss function and fine-tuned GRASP for 22,000 steps using 64 Ascend-910A processors starting from the AFM v.2.3 Model-1 checkpoint. Four inference checkpoints are selected in the fine-tuning process. A multimer prediction model was trained from scratch (Methods) and similarly fine-tuned for 22,000 steps to obtain the fifth model checkpoint to provide diversified checkpoints. In standard GRASP inference, similar to AFM, each of the five models was run with five random seeds. The best predictions were selected based on

predicted local distance difference test (pLDDT) scores and restraint recall. We also applied an iterative noise filtering strategy to reduce the impact of erroneous restraints (Methods).

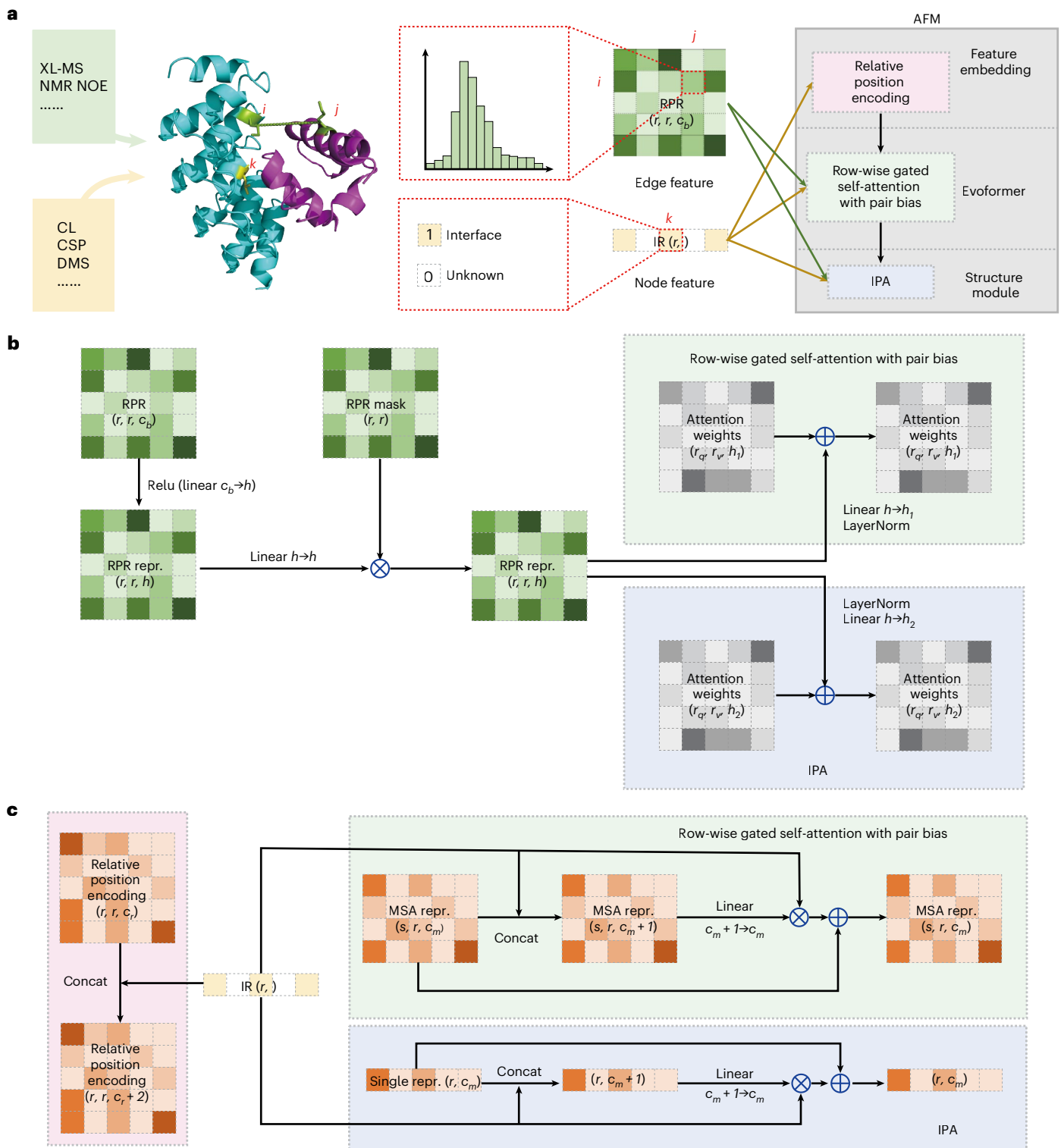
### GRASP effectively uses RPRs and IRs on a self-curated dataset

To evaluate GRASP's performance with the two independent types of restraint, we curated a benchmark dataset by selecting samples released after training set samples. The benchmark set differs from the training set in chain composition, has no internal redundancies, and shows suboptimal AFM predictions (Methods), to prevent data leakage and ensure dataset diversity and quality. This curated dataset contains 713 interfaces from 313 challenging complexes. For RPRs, simulated restraints were sampled from inter-chain residue pairs within 8 Å (contact RPR). The IR restraints were sampled from interface residues whose C $\beta$  atom is within 8 Å of the nearest C $\beta$  atom from the other chain, providing potential contact residues from only one protein. We incorporated varying numbers of simulated restraints and compared the performance of AlphaLink, ClusPro, HADDOCK, ColabDock and GRASP on applicable restraint types, as well as AFM or AF3 without restraints (Supplementary Data).

Without restraints, AF3 outperformed AFM, improving the mean DockQ from 0.17 to 0.23. As the number of restraints increased, the prediction accuracy of the restraint-based methods improved consistently for both restraint types. GRASP consistently ranked first among all methods and delivered the most substantial gains across different quantities of both IRs and contact RPRs (Fig. 2a,b and Supplementary Fig. 1). With just two inter-chain contact RPRs, GRASP reached a mean DockQ of 0.35, with 52.7% of samples exceeding the acceptable threshold of 0.23. For the less informative IRs, GRASP achieved mean DockQ scores and success rates of 0.24 and 35.3%, 0.34 and 51.9%, and 0.41 and 63.2% for 4, 10 and 20 restraints, respectively (Fig. 2a,b and Supplementary Table 1). For the protein complex of Protein Data Bank (PDB) ID 7PBZ (AFM DockQ 0.02), GRASP improved the DockQ to 0.87 using two contact RPRs. Similarly, in the case of PDB 7UBZ, GRASP raised DockQ score from the AFM DockQ of 0.01 to 0.9 with four IRs. In other examples including PDB 7NXX, 7F15 and 7NDF, notable improvements are also observed (Fig. 2c). We note that for both RPR and IR, ColabDock, HADDOCK and ClusPro showed overall better performance than AFM. AlphaLink, trained on 25 Å Sulfo-SDA restraints and not specifically designed to accommodate 8 Å restraints, exhibited limitations in processing restraints of short distances.

The model confidence, pLDDT, exhibited positive Pearson correlations with key performance metrics: DockQ ( $r = 0.39$ ), TM score ( $r = 0.65$ ) and LDDT ( $r = 0.87$ ) (Supplementary Fig. 2a,b). In addition, improvements in pLDDT relative to AFM correlated with corresponding gains in DockQ, TM score and LDDT, yielding Pearson correlations of 0.38, 0.58 and 0.77, respectively (Supplementary Fig. 2c). These findings establish pLDDT as a robust confidence measure, both within GRASP predictions and in comparison to other models. To refine prediction selection, we integrated model and restraint confidence to define a ranking score as  $\text{pLDDT} + 1,000 \times (\text{recall} \geq 0.3)$ . Most samples demonstrated positive intra-sample Spearman correlations between ranking scores and performance for both contact RPR and IR restraints (Supplementary Fig. 2d), underscoring the effectiveness of this combined metric in identifying high-quality predictions. GRASP also yielded reliable pLDDT scores for interface residues when using restraints, with median iPLDDT values of 84.3 and 80.1 for RPRs and IRs, respectively.

To examine whether GRASP reconstructs physically meaningful interfaces, three key interface properties were assessed: binding energy, cross-interface hydrogen bonds and buried solvent-accessible surface area (Supplementary Methods)<sup>34,35</sup>. GRASP-predicted structures exhibited similar binding energy (median  $-23.2$  REU versus  $-21.4$  REU, where 1 REU (Rosetta energy unit)  $\approx 0.4$  kcal mol<sup>-1</sup>; ref. 36), hydrogen bond numbers (median 5 versus 4) and interface areas



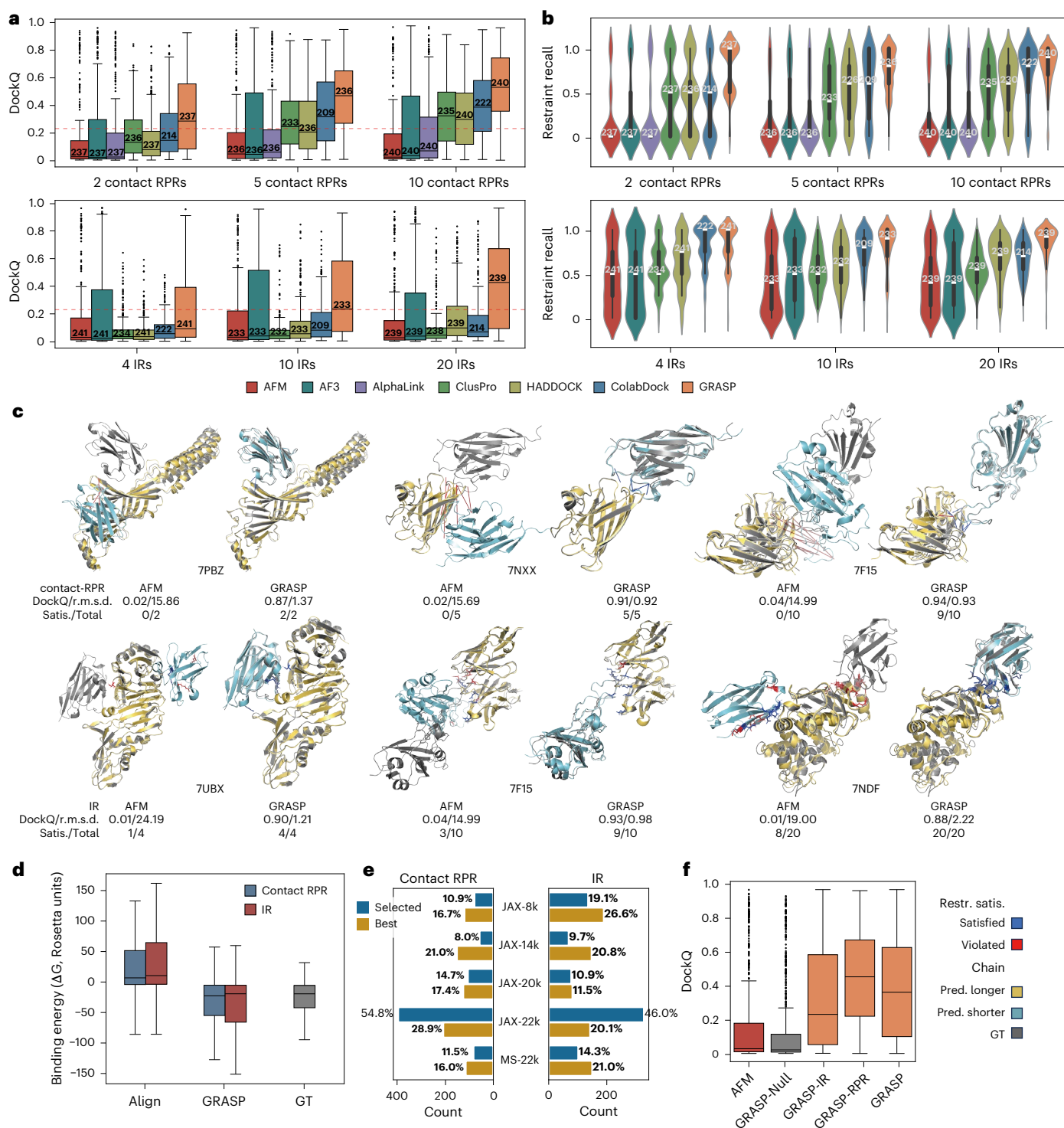
**Fig. 1 | Scheme of GRASP to integrate generalized restraints. a**, The GRASP architecture shows how two types of restraint from various experiments are integrated using three AFM blocks. **b, c**, Details of RPR (**b**) and IR (**c**) integration

are illustrated, with arrows indicating information flow and array shapes noted in parentheses. IPA, invariant point attention; MSA, multiple sequence alignment; NOE, nuclear Overhauser effect; ReLU, rectilinear units; repr., representation.

(median 1,587.7 versus 1,515.7) to the experimental structures (Fig. 2d and Supplementary Fig. 3). In contrast, aligning AF2-predicted monomers to GRASP models without interface remodeling resulted in lowered binding affinity (median 5.9 REU; Fig. 2d). The heavy atom root mean-square deviations (r.m.s.d.s) of interface residues were also lower for GRASP-predicted structures than for AF2-aligned

monomers (one-sided pairwise *t*-test,  $P = 0.003$ ). These observations indicate that GRASP not only positions protein monomers following restraint information, but it also remodels their interfaces, enhancing physical fitness.

We further performed a series of ablation analysis of model number, restraint effect and noise filtering strategy on model performance.



**Fig. 2 | Performance of different methods on benchmark dataset with two types of basic restraint. a, b,** DockQ (a) and restraint recall (b) comparisons across methods for varying numbers of contact RPRs (upper) and IRs (lower). Red dashed lines indicate a DockQ of 0.23. In the boxplots, the center line denotes the median, box edges represent the interquartile range (IQR) and whiskers extend to  $1.5 \times \text{IQR}$ , with points beyond classified as outliers (same for d and f). In violin plots, the white dot marks the median, the thick bar indicates the IQR, the thin line extends to  $1.5 \times \text{IQR}$  and the width reflects data density. Sample sizes are noted above the median lines. ClusPro failed on some samples, and ColabDock could not handle proteins over 1,200 residues, leading to smaller sample sizes. Each restraint type includes 713 cases across three levels. c, Representative structures of GRASP and AFM on selected samples, with DockQ,

r.m.s.d., satisfied (satis.) and total restraints shown. Short predicted chains are cryo, long chains yellow, superimposed on the native structure (gray). All IRs are displayed as sticks, while restrained residue pairs are connected by solid lines. Blue indicates satisfied restraints, while red shows violated ones. AFM was run without restraints, GRASP restraints are also shown on the AFM structures only for comparison. d, Boxplot comparing the binding energy of aligned structures, GRASP predictions and the ground truth (713 samples per box). e, Proportion of the selected structures and the best structures (by DockQ) from the five models in GRASP. f, Ablation study for GRASP performance with and without restraints (restr.). GRASP without restraints (GRASP-Null) showed no improvement and is comparable to AFM. Each box contains 713 samples, except the final GRASP box (GRASP-IR and GRASP-RPR combined) with 1,426 samples. Pred., prediction.

To evaluate the contribution of the five models, we examined their contribution to the final selections and best predictions. Each model contributed to the final selections, with the JAX-22k checkpoint contributing the most. The best structures were more evenly distributed among the models than the selected ones (Fig. 2e). Similar to AFM, using multiple models and selecting the top-scoring prediction enhanced both accuracy and robustness, and reduced the risk of failure (Supplementary Fig. 4).

We also evaluated the inference gain of GRASP from restraint by running an ablation experiment of the default restraint fine-tuned GRASP model without inputting restraints, mimicking the AFM protocol. GRASP without restraint shows comparable accuracy to AFM, with mean DockQs of 0.15 and 0.17, respectively (Fig. 2f). Thus, the improvements by GRASP stem from its ability to incorporate restraints rather than fine-tuning. Furthermore, the chain length slightly negatively correlate with prediction accuracy, showing larger proteins require more restraints for accurate predictions (Supplementary Fig. 5). With no erroneous restraints incorporated, iterative noise filtering had minimal impact on performance (Supplementary Fig. 6).

### GRASP improves prediction using XL, CL and CSP restraints

GRASP and other methods are then benchmarked on XL, CL or CSP data. XL is one of the most common experiments to obtain RPR information. To evaluate the ability of integrating XL data, we simulated three sets of SDA XL restraints on the 294 complexes from the self-curated dataset and collected experimental XL data for additional seven protein complexes. The number of restraints in the three sets was set to 1%, 2% and 5% of the total sequence length, mimicking scenarios of sparse restraints such as in protein interactome, standard level restraints<sup>37</sup>, and a mixture of multiple XL reagents, respectively. In all sets, the maximum restraint distance was set to 25 Å, with 20% of restraints violating the threshold to mimic real-world experimental XL data (Methods). Since ColabDock can only handle restraints below 22 Å, and ClusPro only allows restraints between two rigid chains or subcomplexes, for these three datasets we evaluated GRASP, AlphaLink and HADDOCK.

Overall, GRASP demonstrated superior performance for all restraint coverages, consistently outperforming HADDOCK and AlphaLink in both accuracy and restraint satisfaction (Fig. 3a, Supplementary Table 2 and Supplementary Data). For comparison, AFM achieved a mean DockQ of 0.07. With only 1% restraint coverage, GRASP improved the mean DockQ to 0.18, outperforming both HADDOCK (0.06) and AlphaLink (0.12). GRASP obtained mean DockQ scores of 0.21 and 0.27 at 2% and 5% coverage, respectively, higher than HADDOCK's 0.08 and 0.10, and AlphaLink's 0.13 and 0.20. GRASP also excelled in restraint satisfaction metrics across all coverage levels. For instance, at 2% restraint coverage, GRASP achieved a median satisfaction rate of 69% for all and 81% for correct restraints, outperforming both AlphaLink (37% and 42%) and HADDOCK (61% and 71%). Even with 20% erroneous restraints, GRASP achieved a median satisfaction rate of over 80% for correct restraints, indicating its robustness against experimental noise. Iterative noise filtering consistently improved DockQ and pLDDT across all levels of restraint coverage (Supplementary Fig. 7a). Examples such as PDB 7SQP show notable improvement (DockQ increasing from 0.06 to 0.83; Supplementary Fig. 7b).

GRASP also performs well on large complexes such as PDB 7R94 (>3,500 residues, Fig. 3b). In this case, AFM mispositioned many chains, and even after adding simulated XLs, HADDOCK cannot rescue the prediction. Both AlphaLink and, especially, GRASP showed marked improvements, yielding DockQs of 0.51 and 0.67, respectively.

We gathered experimental data from various studies (Methods and Supplementary Data) for further tests. We collected nine samples with XL data<sup>21,38–41</sup> (Fig. 3c left). GRASP achieved a mean DockQ of 0.48, outperforming other tools (AFM, 0.31; ClusPro, 0.05; HADDOCK, 0.05; AlphaLink, 0.38) (Supplementary Table 3 and Supplementary Data). For the four non-antigen–antibody complexes, GRASP, AlphaLink and

AFM made predictions with DockQ higher than 0.23. Among the five antibody–antigen XL examples, AFM had only one prediction with DockQ > 0.23 but with a large r.m.s.d. of 26.37 (Supplementary Fig. 8). After incorporating experimental XL data, GRASP showed improvements in 80% of the cases. For PDB 4G3Y, using 12 DSBUs (30 Å crosslink distance) restraints including three erroneous ones, GRASP increased the DockQ from 0.03 (AFM) to 0.77, satisfying all nine correct restraints while ignoring the three incorrect ones. AlphaLink also predicted the correct structure with DockQ of 0.67. In contrast, HADDOCK was biased toward incorrect restraints, resulting in a DockQ below 0.23 (Fig. 3d, upper). Therefore, GRASP not only ensures structure alignment with the provided restraints, but also excels in restraint discrimination, minimizing the impact of erroneous restraints.

For five cases with experimental CL data<sup>28</sup>, AFM achieved an average DockQ of 0.45, while HADDOCK and ClusPro yielded average DockQs of 0.3 and 0.04, respectively. ColabDock improved to an average DockQ of 0.56, and GRASP showed the greatest improvement, reaching an average DockQ of 0.58 (Fig. 3c). Notably, for 4INS8, GRASP (DockQ of 0.56) is the only method that outperforms AFM (DockQ of 0.43) (Fig. 3d).

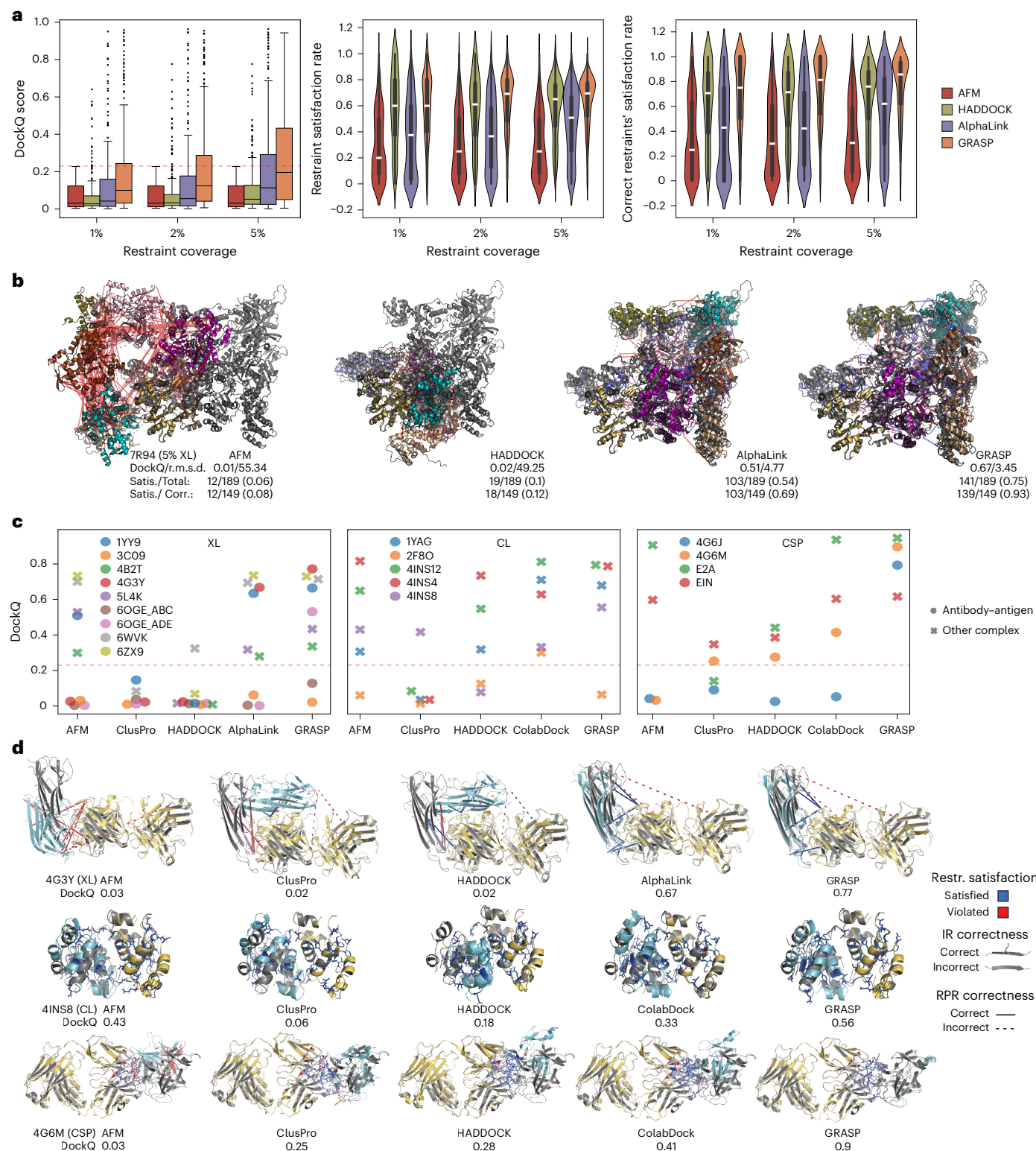
For the four CSP cases from two studies<sup>24,42</sup> (Fig. 3c, right), AFM performed well in two cases but struggled with the two antibody–antigen complexes, yielding an average DockQ of 0.39. HADDOCK and ClusPro showed average DockQs of 0.28 and 0.21, respectively. ColabDock improved this value to 0.5, while GRASP further raised it to 0.81. For instance, GRASP accurately predicted PDB 4G6J with a DockQ of 0.79, whereas all other methods failed. In another challenging case of PDB 4G6M, GRASP improved the DockQ to 0.9 using CSP data. ClusPro, HADDOCK and ColabDock also improved but only reached acceptable DockQs (Fig. 3d). Notably, all methods except GRASP struggled with these antibody–antigen complexes, highlighting the challenge of this task (Fig. 3c,d).

### GRASP performance on antibody–antigen prediction

Accurate structure prediction of antigen–antibody complexes is critical for antibody design and immunotherapy. However, current methods such as AFM have limited accuracy. AF3 was reported to perform better on such complexes<sup>5</sup>, and is included in our evaluation in this section.

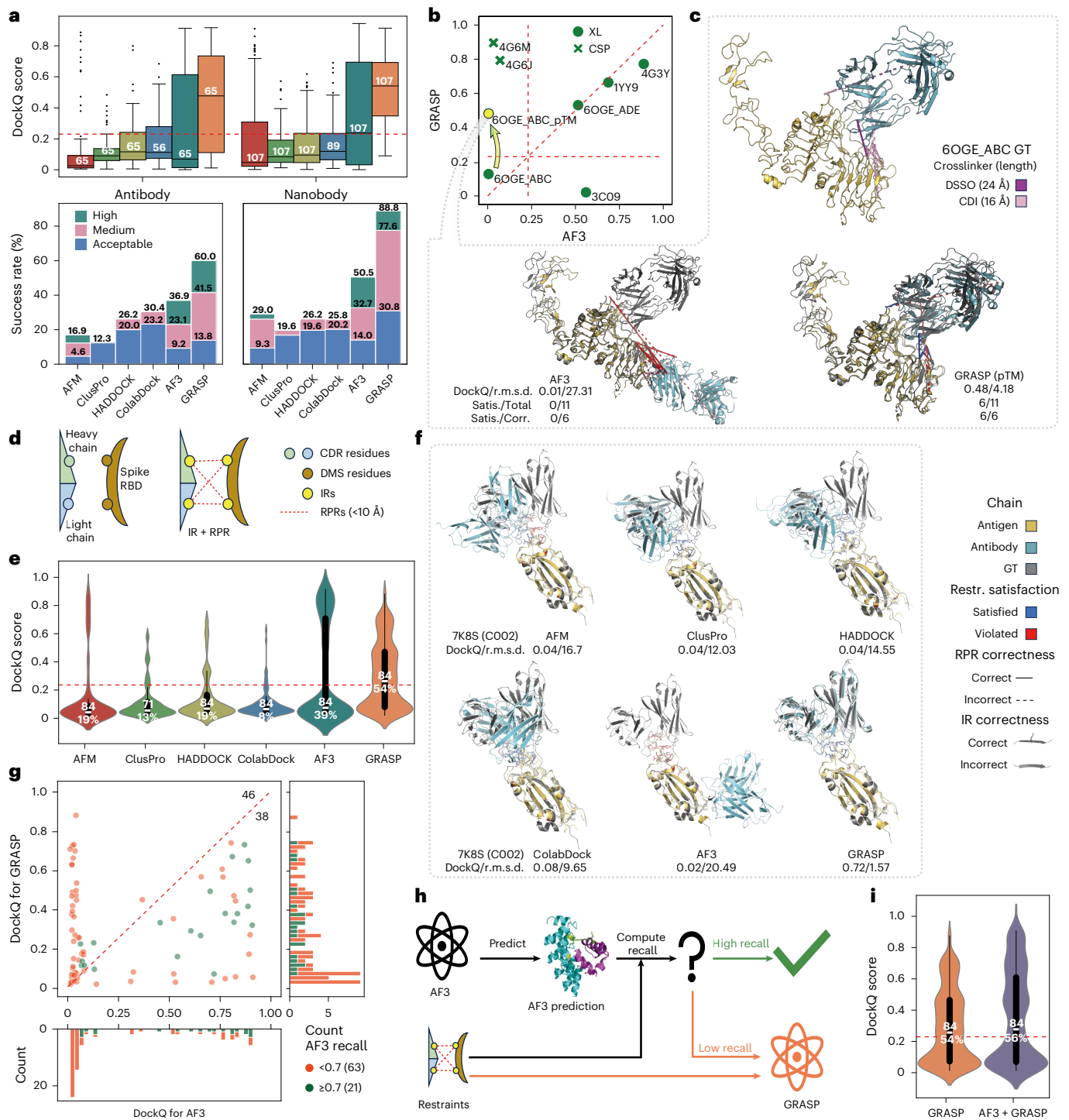
We first evaluated GRASP on the BM5.5 (ref. 43) antibody–antigen dataset, which comprises 67 samples, each with three sets of simulated DMS IRs (Methods). GRASP surpassed all other methods with a DockQ median of 0.64 and a success rate of 71.6% (Supplementary Fig. 9, Supplementary Table 4 and Supplementary Data). To avoid data leakage, we examined GRASP performance with an additional set of DMS restraints generated for cases curated in ref. 44, comprising 65 antibodies and 107 nanobodies released posttraining (Methods). GRASP yielded median DockQ scores of 0.477 for antibodies and 0.541 for nanobodies with success rates of 60.0% and 88.8%, respectively, substantially surpassing the second-best method, AF3 (median DockQ 0.069 and 0.237, success rate 36.9% and 50.5%, respectively) (Fig. 4a, Supplementary Fig. 10 and Supplementary Table 5). Other docking methods all exhibited lower success rates. These findings underscore GRASP's superiority over both deep learning-based (AF3) and restraint-guided docking approaches.

We further evaluated GRASP assisted by experimental data, including 7 CSP or XL samples from the previous section and 84 additional DMS samples. GRASP achieved a success rate of 5 out of 7, outperforming AF3's 4 out of 7 (Fig. 4b and Supplementary Data). In one of GRASP's two failure cases, PDB ID 3C09 (Supplementary Fig. 11), 10 of 12 experimental restraints were not satisfied by the X-ray structure, suggesting a possible alternative binding conformation. In the second case PDB 6OGE\_ABC, with 5 of 11 XLs unsatisfied by experimentally resolved reference structure, GRASP achieved a DockQ of 0.13, outperforming AF3's 0.01 and AlphaLink's 0.004 (Fig. 4c and Supplementary Fig. 12). Notably, GRASP predicted the correct structure with a DockQ score of 0.48, which failed to be identified by the pLDDT-based ranking score



**Fig. 3 | The performance of GRASP and other methods using single-type experimental restraint data. a**, DockQ (left), restraint satisfaction rate (middle) and correct restraint satisfaction rate (right) for various methods on simulated SDA XL-MS data across different restraint coverages. In boxplots, the center line denotes the median, box edges represent the IQR and whiskers extend to  $1.5 \times$  IQR, with points beyond classified as outliers. In violin plots, the white dot marks the median, the thick bar indicates the IQR, the thin line extends to  $1.5 \times$  IQR and the width reflects data density. Each box and violin includes 294 samples. The red dashed horizontal line in the left subfigure indicates a DockQ value of 0.23 (same for subfigures in c). **b**, 3D visualization of predicted structures by various methods compared to the true structure (gray) on 7R94 in the simulated XL dataset. **c**, DockQ

scores for various methods using single-type real-world experimental restraint data, including XL (left), CL (middle) and CSP (right). Since the number of cases for each experimental approach is lower than ten, all of the evaluated cases are drawn separately as scatters for different methods. **d**, 3D cartoons of representative predicted and true structures for each method under different experimental restraints. The predicted structures are colored cyan (smaller part) and yellow (larger part), with the latter superimposed on the real structure in gray. Satisfied restraints are shown in blue and violated ones in red. Correct IRs are displayed as sticks, incorrect ones as cartoons. Correct RPRs are connected by solid lines, while incorrect RPRs are shown with dashed lines. AFM was used without restraints, while GRASP restraints are displayed on the AFM structure for comparison.



**Fig. 4 | GRASP improves antigen–antibody structure prediction accuracy.**

**a**, Performance on the simulated DMS Hitawala–Gray dataset for antibodies (left) and nanobodies (right). Top, boxplots of DockQ distributions. The center line denotes the median, box edges represent the IQR and whiskers extend to 1.5× IQR, with points beyond classified as outliers. Sample sizes are noted above the medians. Bottom, stacked bar plots of success rates at three accuracy levels for antibody (left) and nanobody (right). Numbers above bars indicate cumulative percentages. **b**, Scatter plot comparing GRASP and AF3 DockQs for antigen–antibody complexes with XL or CSP restraints. The yellow dot highlights the case in **c**. **c**, 3D visualization of GRASP (pTM-based) and AF3 predictions for 60GE\_ABC. Top, two types of crosslinker on the ground truth (GT) structure. Bottom, predicted structures by AF3 and GRASP (antibody, cyan; antigen, yellow), with antigen superimposed on GT (gray). Satisfied restraints are colored blue, and violations red (same in **f**). Correct RPRs appear as solid lines; incorrect

as dashed lines. **d**, Schematic of GRASP’s restraint strategy on the experimental COVID-19 DMS dataset. **e**, Violin plot of DockQs on the experimental DMS dataset. The white dot marks the median, the thick bar shows the IQR, the thin line extends to 1.5× IQR; numbers above and below the median indicate case counts and success rates, respectively (same in **i**). **f**, 3D predictions on PDB 7K8S (antibody C002). DMS sites are shown, with correct sites shown as sticks. **g**, Scatter plot comparing AF3 and GRASP DockQs on samples with high (orange) and low (green) AF3 DMS recall. Numbers beside  $y = x$  line indicate sample counts on each side. Parentheses in the legend show the total samples per category. **h**, Schematic of the strategy combining AF3 and GRASP. **i**, Violin plot of DockQ for the combined strategy versus GRASP alone. The slanted, vertical and horizontal red dashed lines in relevant subfigures indicate  $y = x$ ,  $x = 0.23$  and  $y = 0.23$ , respectively. DSSO, disuccinimidyl sulfoxide; CDI, carbonyldiimidazole.

but could be selected by a pTM-based score ( $0.8 \times \text{ipTM} + 0.2 \times \text{pTM}$ ), suggesting that adjusting the ranking strategy can further enhance performance in certain cases (Fig. 4b,c). Under both ranking schemes, GRASP consistently met all correct restraints and ignored the incorrect ones, again demonstrating its resistance to experimental noise.

We also compiled a new DMS dataset from experiment<sup>45</sup>, containing 84 different monoclonal antibodies bound to the Receptor Binding Domain (RBD) of the SARS-CoV-2 spike protein (Methods). We selected residues with escape scores above 0.2 as the antigen-side interface to reduce noise (Supplementary Fig. 13). We used complementarity determining region (CDR) residues as the antibody-side interface and also applied RPRs across the interface residues on both sides (Fig. 4d). GRASP outperformed all other methods with a DockQ median of 0.25 and a success rate of 53.6%. In comparison, AF3 has a DockQ median of 0.07 and a success rate of 39.3% (Fig. 4e, Supplementary Table 6 and Supplementary Data). For PDB 7K8S (Fig. 4f), AFM and AF3 performed poorly, and AF3 misidentified the binding site entirely. ClusPro, HADDOCK and ColabDock met DMS restraints but failed to predict the correct binding orientation. GRASP corrected the interface in assistance of DMS data and accurately predicted the structure with a DockQ of 0.72.

Notably, AF3 excelled in some cases (Fig. 4g), outperforming GRASP in 38 out of 84 cases, 16 of which showed over 70% agreement with DMS restraints. DockQ gains of GRASP over AF3 correlate positively with information gain from restraints (Pearson  $r = 0.45$ ), defined as the difference between extra information and erroneous information provided by restraints (Supplementary Fig. 14). To take advantage of both GRASP and AF3, we proposed a joint protocol (Fig. 4h): to use AF3 for initial predictions and accept structures that align well with experimental restraints, otherwise use GRASP to improve predictions by restraints. This approach improved the overall prediction to a median DockQ of 0.28 and a success rate of 56.0%, surpassing both GRASP (0.25, 53.6%) and AF3 (0.06, 39.3%) (Fig. 4e,i and Supplementary Table 6).

Antigen–antibody complex prediction is challenging due to CDR variability and absent coevolutionary signals<sup>46</sup>, limiting MSA-based models such as AFM and AF3. GRASP improves the performance by effectively integrating diverse external restraints (for example, XL, DMS, CSP and CDR), which constrain the conformational search and guide predictions toward native-like binding. Its success underscores the value of experimental and prior knowledge in antibody modeling.

### GRASP serves as a toolkit for integrative modeling

Single experimental technique often yields limited structural information, sometimes insufficient to fully determine protein interaction. Integrating multi-source experimental restraints to assist modeling is a promising way to study the structure and function of protein complexes<sup>47</sup>. To extend GRASP's applicability, we further expand its usable restraint types to include cryo-electron microscopy (cryo-EM) density map. We iteratively dock chain combinations of the GRASP-predicted structure into the density map using a 'random-split' strategy with our graphical processing unit (GPU)-accelerated program Combfit, and then extract the spatial arrangement of subunits as RPRs to guide GRASP prediction (Supplementary Fig. 15 and Methods). Since a high-resolution density map can resolve atomic structures with specialized software packages such as ModelAngelo<sup>48</sup>, we focus here on low-to-medium resolution maps. We evaluate GRASP's integrative modeling performance across diverse combinations of experimental approaches (Supplementary Methods).

The first case contains experimental information from XL, mutagenesis and the density map. It characterizes human APOBEC3G (A3G) proteins bound to the substrate receptor module of CRL5 containing HIV-1 Vif, CBF $\beta$ , Elongin B and Elongin C (VCBC). We first applied the mutagenesis data between A3G and Vif and XL-MS data containing 13 pairs of crosslinked peptides<sup>49</sup> to model this system. Nine out of 13 crosslinked residue pairs satisfy the maximum Ca–Ca distance of 24 Å in

the ground truth structure (PDB 8CX0) (Fig. 5a). AFM mispredicted the A3G-Vif binding interface, violating six XLs. When integrating XL and mutagenesis data, GRASP satisfied all IRs and 10 of 13 XLs with a DockQ of 0.398, but A3G still took a wrong orientation. By incorporating a medium-resolution cryo-EM density map of 7.82 Å, downsampled from the original map (EMD-27032), GRASP performance further improved, yielding a correct binding pose of Vif and a higher quality interface (DockQ 0.552 versus 0.307 for IMP).

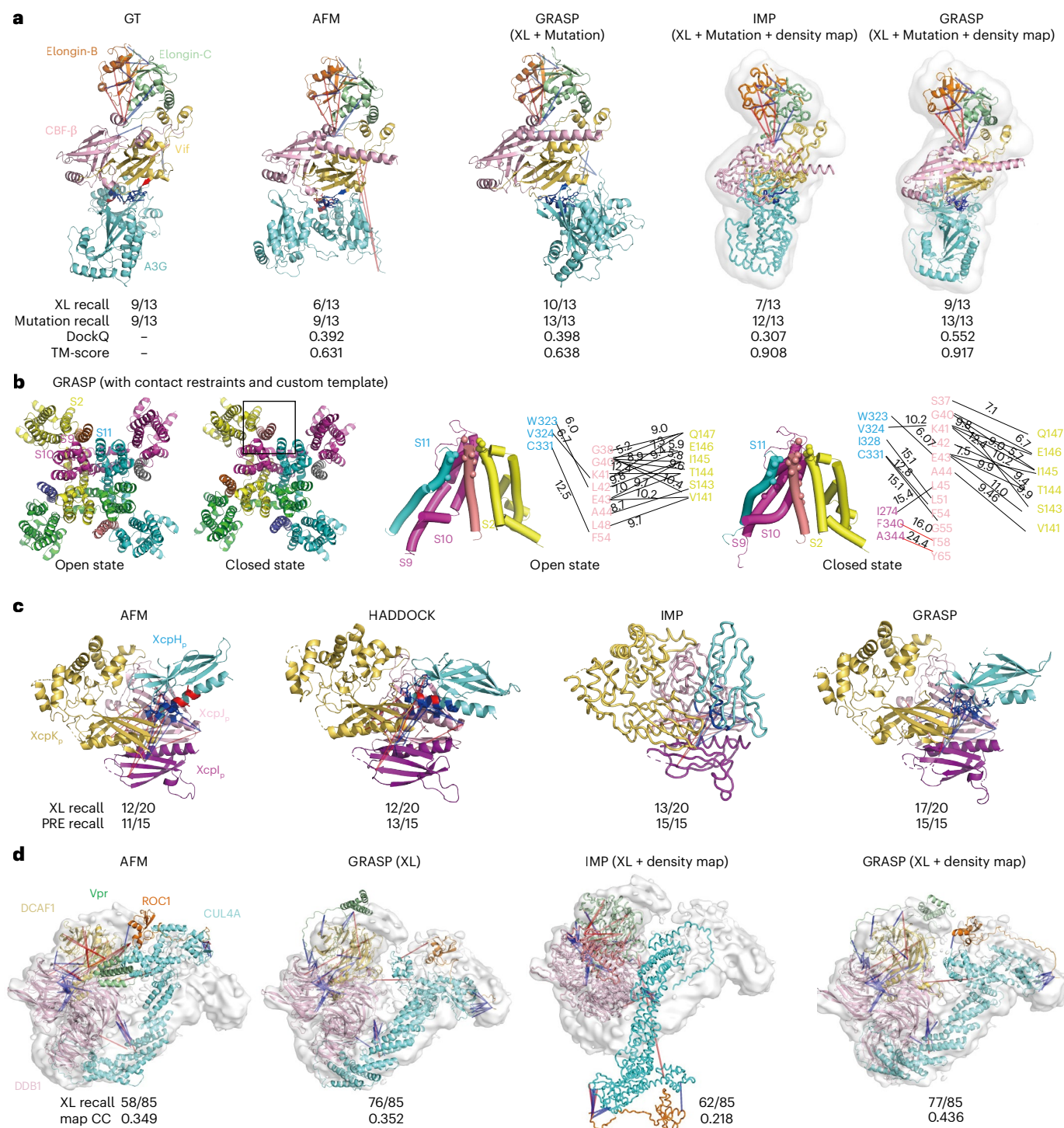
In the second case, we used experimental information from disulfide XL, Cd(II)-cysteine bridging and double mutant cycle analysis from ref. 50 to model the open and closed state of voltage-gated potassium (K) channel KCNQ1 and its regulatory protein KCNE1 with a stoichiometry of 4:4. These restraints were organized in a state-dependent manner and unified into RPRs. The structures of KCNQ1 from the KCNQ1–KCNE3 complex<sup>51</sup> in an open or closed state were used as single-chain templates to bias GRASP's prediction toward the corresponding state. Notably, AFM-predicted structures are similar to the closed state with low confidence on the position of KCNE1 (Supplementary Fig. 16a). Integrating multi-source RPRs and using custom single-chain templates, GRASP successfully predicted the binding interface in both states (Fig. 5b). IMP failed to keep the KCNE1 to the binding cavity formed by KCNQ1 (Supplementary Fig. 16b). The GRASP-generated structures satisfied most of the restraints in both states except two in the closed state. Models built by the group in ref. 50 using Rosetta protein–protein docking satisfied the same restraints with GRASP in both states (Supplementary Fig. 16c–e). Compared with them, GRASP predicts smaller motion amplitudes in the peripheral domain between open and closed conformations.

In the third case, we modeled the type II secretion system, where XcpH is assembled into the complex XcpIJK, using XL data and NMR paramagnetic relaxation enhancement (PRE) data from ref. 52. The AFM-predicted structure satisfied 15 of 20 XL and 11 of 15 PRE restraints with one end of the N-terminal helix of XcpH positioned far from XcpIJK (Fig. 5c). One research group<sup>52</sup> built a model using HADDOCK with the XcpH helix in parallel to the XcpJ helix, satisfying more PRE restraints. IMP predicted a completely different pose of XcpH but it satisfies more restraints than that predicted by HADDOCK. The GRASP-built structure is similar to the HADDOCK model but the helix of XcpH is slightly removed from XcpIJK, with 18 of 20 XLs and all 15 PRE restraints satisfied.

The fourth modeled system is a 5-subunit CRL4<sup>DCAF1-Ctd</sup>/Vpr<sub>mus</sub> assembly in state-1 integrating XL data and the density map (EMD-10612) from ref. 53. The accessory viral protein Vpr of the simian immunodeficiency virus can redirect the specificity of host Cullin4-RING ubiquitin ligases to the antiviral host factor SAMHD1. The AFM-predicted structure conflicted with the density map, placing the CUL4A head close to other subunits and satisfying only 58 out of 85 XLs (Fig. 5d, left panel). By integrating XLs alone, GRASP-predicted structure satisfied 76 XLs and obtained a cross-correlation of 0.352 with the predicted head of CUL4A fitting into the density map (Fig. 5d, middle left panel). By integrating both XL and density map, GRASP further improved to 77 satisfied XLs and a cross-correlation of 0.436, outperforming the former two models. In contrast, the IMP-predicted structure satisfied 62 XLs with a lower density map cross-correlation of 0.218 (Fig. 5d, right two panels).

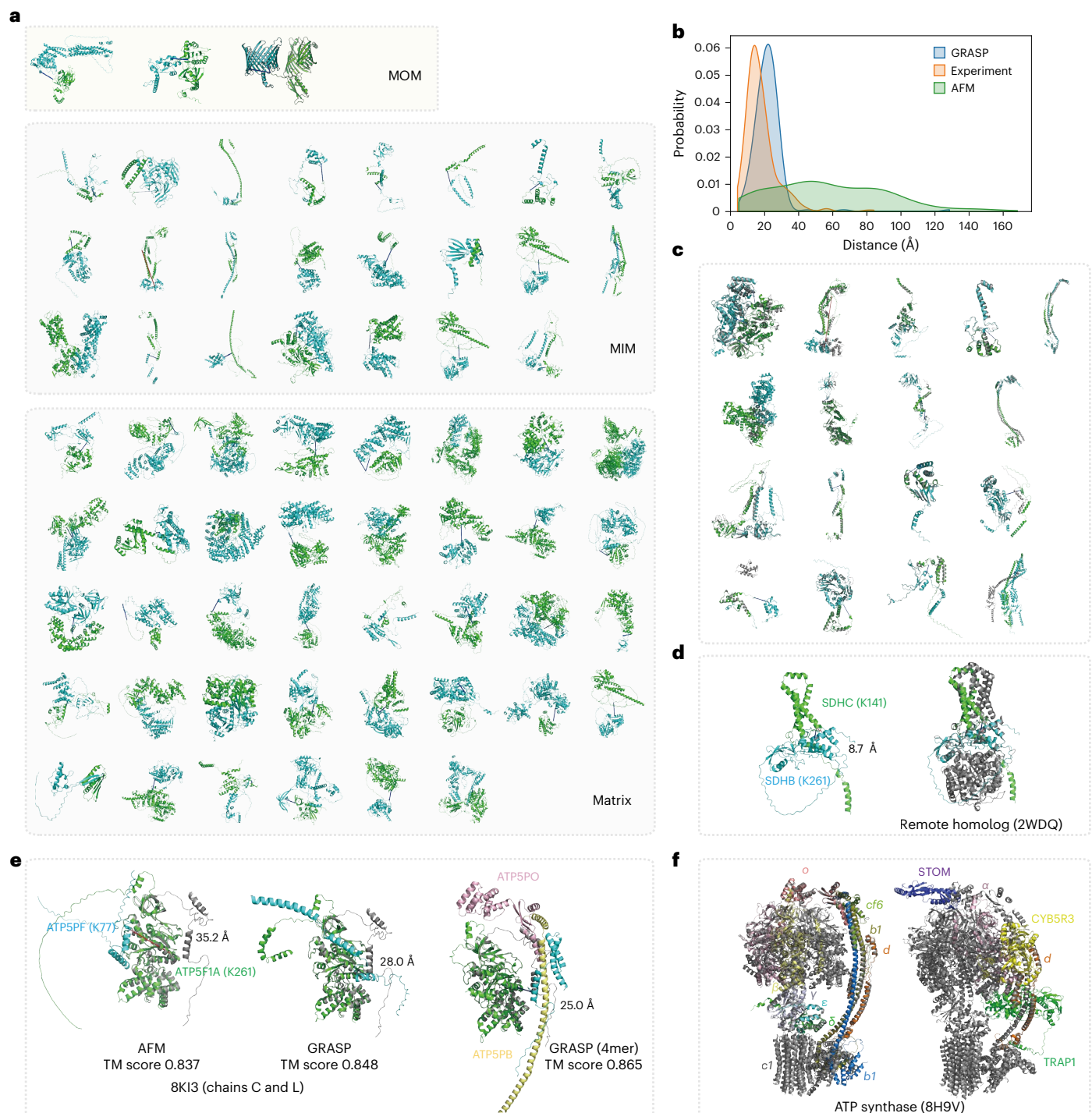
### Structural insights into in situ mitochondrial PPI

Traditional structure determination methods, such as X-ray crystallography and cryo-EM, require purified protein complexes and stable subunit interactions, making it difficult to capture transient and dynamic PPIs. Recently, improvements in chemical XL techniques have enabled the detection of protein interactomes, even at the whole-cell scale<sup>37,54</sup>. While in-cell XL-MS reveals valuable in situ interaction information, it lacks atomic details. Here, we used GRASP to build the corresponding structure models.



**Fig. 5 | Applying GRASP in integrative modeling results in reasonable structures.** **a**, Visualization of experimental restraints for A3G-VCBC complex on ground truth (PDB ID 8CX0, left), AFM-predicted structure (middle left), GRASP-predicted structure using XL and mutagenesis (middle), IMP-predicted structure and GRASP-predicted structure using XL, mutagenesis and density map (middle right, right). Blue lines refer to satisfied XL restraints and red lines refer to violated ones (same in **c** and **d**). XL recall refers to the ratio of crosslinked residue pairs with a Ca–Ca distance below 24 Å in **a**, and below 25 Å in **c** and **d** (Supplementary Table 7). Mutagenesis recall indicates ratio of restrained residues whose C $\beta$  atoms are within 10 Å of the C $\beta$  atoms of residues from another chain. **b**, Predicted structures for the KCNE1–KCNE1 complex from GRASP with contact restraints and custom templates (left two panels). S1–S11 refer to the helices arranged from N terminal to C terminal. The right two panels

are enlarged interface labeled in left middle panel. The Ca–Ca distance computed for each the used RPRs are labeled. Two violated restraints (F340/KCNE1-T58/KCNE1; A344/KCNE1-Y65/KCNE1) are colored in red. Compared with the open state, the KCNE1 S11 tail in the closed state twists toward the ion channel center and more contacts exist between KCNE1 and the KCNE1 S9 or S10. **c**, Visualization of modeling structure of XcpHJK from AFM, HADDOCK, IMP and GRASP using XL and PRE data. PRE recall is the ratio of restrained residues whose C $\beta$  atoms are within 10 Å of the C $\beta$  atoms of residues from another chain. **d**, Visualization of experimental restraints for AFM-predicted structure (left), GRASP-predicted structure using XL (middle left), IMP-predicted structure (middle right) and GRASP-predicted structure (right) using both XL and the density map. Map cross-correlation measures the degree of agreement between the structure and the density map.



**Fig. 6 | Dimeric structure models are predicted by GRASP for PPIs captured by targets XL-MS in mitochondria. a**, Visualization for PPIs colocalized in the same subcellular localization for mitochondria outer membrane (MOM, upper), mitochondria inner membrane (MIM, middle) and matrix (lower) with pLDDT >75. Blue lines refer to the well-satisfied restraints and red lines refer to violated restraints. **b**,  $C\beta$ - $C\beta$  distance distribution of crosslinked residues in the GRASP-predicted structure, AFM-predicted structure and experimental statistics. **c**, Mapping of 17 pair of GRASP-predicted interactions on ground truth. Ground truths are colored in gray. **d**, Structure model of the SDHB-SDHC interaction. Blue lines refer to the well-satisfied restraints and red lines refer to violated restraints. The predicted structure is aligned with a homologous protein (PDB 2WDQ) in

the right panel with the ground truth structure colored in gray. **e**, Visualization of prediction structures for ATP5F1A-ATP5PF by AFM, GRASP (left two panels). GRASP-predicted four-body interaction including extra two sequences are shown in the right panel. Distances of crosslinked residues are displayed around each panel. The ground truth structure (chains C and L of PDB 8IK3) is colored in gray. **f**, Visualization of PPIs related to FO-F1 ATP synthase. PPIs within ATP synthase are shown on the left panel and representative predicted PPIs between ATP synthase and other proteins are shown on the right panel. The ground truth structure of ATP synthase (PDB 8H9V) was colored in gray and crosslinked chains in ATP synthase are in different colors.

In the following example, we collected XL-MS data from ref. 55 in which a targeted delivery system was developed to map protein interactions in mitochondria in situ. A total of 158 inter-protein crosslinks were mapped to 144 PPIs using the human MitoCarta v.3.0 database<sup>56</sup>. We transformed these crosslinked peptides into RPRs with an upper distance limit of 30 Å to predict protein interactions. Among the 144 predicted PPI pairs, 121 showed pLDDT above 75 (Fig. 6a and Supplementary Fig. 17). In addition, 29 out of 69 predicted PPIs with a direct interaction exhibited an interface pLDDT above 75 (Supplementary Data). Compared to AFM, the addition of XLs into GRASP leads to structure adjustment to varying degrees (Supplementary Fig. 18a and Supplementary Data). GRASP satisfied XLs in 140 out of 144 predicted PPIs, whereas AFM satisfied only 31. Further, the distance distribution of GRASP's predicted crosslinked residues closely matched the experimental statistics<sup>57</sup> of reagent disuccinimidyl suberate while AFM predictions resembled a uniform distribution (Fig. 6b).

We then validated the predictions by mapping them to available ground truth structures (Fig. 6c). Experimentally resolved PPI structures in mitochondria are very limited and only 17 pairs were identified from the Protein Data Bank. GRASP accurately predicted most PPIs with a median TM score of 0.881 (0.838 for AFM; Supplementary Fig. 18b), and exhibited binding energies and interface sizes similar to the native ones (Supplementary Fig. 18c,d). The SDHB–SDHC interaction, involved in the tricarboxylic acid cycle, was modeled by GRASP with K141 of SDHC close to K261 of SDHB, satisfying the XL restraint with a Ca–Ca distance of 8.7 Å (Fig. 6d left). Despite the absence of ground truth structure for this interaction, we identified a pair of remote homologs from *Escherichia coli* (PDB 2WDQ) and observed a highly conserved interface (TM score 0.916, Fig. 6d, right).

In some cases, we observed that crosslinked protein pairs lacked direct interaction (Fig. 6c), suggesting the presence of additional binding partners mediating these interactions. Incorporating such mediators may improve structural prediction. For example, ATP5PF–ATP5F1A is a part of the 16-subunit FO–F1 ATP synthase. In modeling the ATP5PF–ATP5F1A dimer, AFM predicted an incorrect orientation of ATP5F1A, while GRASP provided a more reasonable orientation but failed to predict the precise pose (Fig. 6e, left two panels). On including two additional interacting subunits, ATP5PB and ATP5PO, in the GRASP modeling, the relative position of ATP5PF and ATP5F1A improved with a TM score of 0.865 (Fig. 6e, right panel).

FO–F1 ATP synthase (complex V) is the most crosslinked complex in the experiment, with 11 intra-complex restraints and 11 additional restraints linking it to other proteins. Although we only modeled two-body interactions, 6 out of 7 PPIs aligned well with the reference structure across the  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\epsilon$  and  $o$  subunits in F1 and b1, d, cf6 subunits in the rod (Fig. 6d). In addition, we built three interfaces between other proteins and the mature FO–F1 ATP synthase. TRAP1 binds to a d subunit, while STOM and CB5R3 bind to an  $\alpha$  subunit at different sites. All three complexes exhibited minimal clash with the existing ATP synthase, indicating the ability of GRASP to model transient protein interactions. The relative orientation between TRAP1 and d subunit is consistent with the location of TRAP1 in the matrix. For STOM and CB5R3 located in the inner membrane, their predicted orientations relative to ATP synthase also align with their location. TRAP1 is believed to function by balancing the oxidative phosphorylation and aerobic glycolysis<sup>58</sup>, while STOM may influence the activity of ion channel according to Gene Ontology annotation<sup>59</sup>, and CYB5R3 serves as NADH-cytochrome b5 reductase<sup>60</sup>. Associating the predicted interactions with specific function may offer new insights on the regulatory mechanism.

## Discussion

In this work we propose GRASP, a generalized model that integrates a variety of experimental restraints to enhance protein complex structure prediction. By incorporating RPR and IR information directly into

the GRASP model and fine-tuning both the AFM and the self-trained MEGAFold-Multimer checkpoints with additional restraint-related loss terms, GRASP can simultaneously process one or more types of experimental data as restraints to improve the accuracy and reliability of a predicted protein complex structure. This approach allows the integration of multimodal data to complement and/or cross-validate each other. An iterative noise filtering process is also introduced to reduce the impact of erroneous restraints. GRASP infers structure directly and only needs one-time training to obtain an affordable inference time (Supplementary Fig. 19 and Supplementary Data) and AFM comparable GPU usage (Supplementary Fig. 20), supporting diverse restraint forms especially for large complexes.

GRASP supports both RPRs at various distances and IRs. It demonstrates excellent performance in complex structure prediction by effectively integrating various simulated and experimental restraint data. Under the simulated XL or real-world XL, CL and CSP restraints, GRASP achieves higher prediction accuracy than other methods, demonstrating its efficiency in integrating independent data types. For the crucial but highly challenging antigen–antibody prediction task, GRASP leverages sparse restraints from CSP, XL or DMS data to surpass AF3 in accuracy. A combined strategy of GRASP with AF3 further improves prediction. Specifically, the performance advantage of GRASP over AlphaLink may arise from several methodological reasons, including the integration of varied RPRs and IRs across all Evoformer layers and invariant point attention modules, additional restraint-specific loss terms, iterative noise filtering and ensemble prediction from five diverse checkpoints—collectively enabling effective usage of restraints for complex structure prediction.

GRASP naturally serves as an integrative modeling tool. By combining varied information across cryo-EM, XL, NMR PRE, mutagenesis and Cd(II)-cysteine bridging experiments in four cases, we demonstrated the potential for GRASP to perform complicated restrained modeling tasks. With this tool, we are able to model structures with actively developing in situ or dynamic detection techniques, such as in-cell XL, solution NMR and in situ cryo-electron tomography, which helps to reveal potential transient interaction structures and molecular mechanisms near physiological conditions.

For now, GRASP usage is limited to information formats of pairwise distance (RPRs) and IRs. Preliminary attempts to convert cryo-EM density maps into RPRs in combination with Combfit have shown promise. Directly integrating data from methods such as small angle X-ray scattering and cryo-EM will further broaden the applicability of this method.

In the post-AlphaFold stage, structural biologists are focusing on more complex protein interactions with essential biological applications. GRASP can act as a new toolkit for integrating experimental information with different levels of structural details and accuracy. We are optimizing the integrative strategy of GRASP to enable solution of more challenging application scenarios. Moreover, GRASP can also be consolidated into the workflow of existing integrative modeling tools such as CombFold<sup>61</sup>, thus combining the advantages of sampling and/or searching-based methods and deep learning-based methods.

## Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41592-025-02820-1>.

## References

- Pearce, R. & Zhang, Y. Toward the solution of the protein structure prediction problem. *J. Biol. Chem.* **297**, 100870 (2021).
- UniProt, Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Res.* **51**, D523–D531 (2023).

3. Berman, H. M. et al. The Protein Data Bank. *Nucleic Acids Res.* **28**, 235–242 (2000).
4. Evans, R. et al. Protein complex prediction with AlphaFold-Multimer. Preprint at *bioRxiv* <https://doi.org/10.1101/2021.10.04.463034> (2022).
5. Abramson, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500 (2024).
6. Lin, Z. et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **379**, 1123–1130 (2023).
7. Wu, R. et al. High-resolution de novo structure prediction from primary sequence. Preprint at *bioRxiv* <https://doi.org/10.1101/2022.07.21.500999> (2022).
8. Bryant, P. Deep learning for protein complex structure prediction. *Curr. Opin. Struct. Biol.* **79**, 102529 (2023).
9. Graziadei, A. & Rappsilber, J. Leveraging crosslinking mass spectrometry in structural and cell biology. *Structure* **30**, 37–54 (2022).
10. Leitner, A., Faini, M., Stengel, F. & Aebersold, R. Crosslinking and mass spectrometry: an integrated technology to understand the structure and function of molecular machines. *Trends Biochem. Sci.* **41**, 20–32 (2016).
11. Vogeli, B. The nuclear Overhauser effect from a quantitative perspective. *Prog. Nucl. Magn. Reson. Spectrosc.* **78**, 1–46 (2014).
12. Limpikirati, P., Liu, T. & Vachet, R. W. Covalent labeling-mass spectrometry with non-specific reagents for studying protein structure and interactions. *Methods* **144**, 79–93 (2018).
13. Williamson, M. P. Using chemical shift perturbation to characterise ligand binding. *Prog. Nucl. Magn. Reson. Spectrosc.* **73**, 1–16 (2013).
14. Nerli, S., McShan, A. C. & Sgourakis, N. G. Chemical shift-based methods in NMR structure determination. *Prog. Nucl. Magn. Reson. Spectrosc.* **106–107**, 1–25 (2018).
15. Fowler, D. M. & Fields, S. Deep mutational scanning: a new style of protein science. *Nat. Methods* **11**, 801–807 (2014).
16. Starr, T. N. et al. Deep mutational scanning of SARS-CoV-2 receptor binding domain reveals constraints on folding and ACE2 binding. *Cell* **182**, 1295–1310.e1220 (2020).
17. Berger, C. et al. Cryo-electron tomography on focused ion beam lamellae transforms structural cell biology. *Nat. Methods* **20**, 499–511 (2023).
18. Puthenveetil, R. & Vinogradova, O. Solution NMR: a powerful tool for structural and functional studies of membrane proteins in reconstituted environments. *J. Biol. Chem.* **294**, 15914–15931 (2019).
19. Chavez, J. D. et al. Systems structural biology measurements by in vivo cross-linking with mass spectrometry. *Nat. Protoc.* **14**, 2318–2343 (2019).
20. Stahl, K., Graziadei, A., Dau, T., Brock, O. & Rappsilber, J. Protein structure prediction with in-cell photo-crosslinking mass spectrometry and deep learning. *Nat. Biotechnol.* **41**, 1810–1819 (2023).
21. Stahl, K. et al. Modelling protein complexes with crosslinking mass spectrometry and deep learning. *Nat. Commun.* **15**, 7866 (2024).
22. Feng, S. et al. Integrated structure prediction of protein–protein docking with experimental restraints using ColabDock. *Nat. Mach. Intell.* **6**, 924–935 (2024).
23. de Vries, S. J., van Dijk, M. & Bonvin, A. M. The HADDOCK web server for data-driven biomolecular docking. *Nat. Protoc.* **5**, 883–897 (2010).
24. Dominguez, C., Boelens, R. & Bonvin, A. M. J. J. HADDOCK: a protein–protein docking approach based on biochemical or biophysical information. *J. Am. Chem. Soc.* **125**, 1731–1737 (2003).
25. Kozakov, D. et al. The ClusPro web server for protein–protein docking. *Nat. Protoc.* **12**, 255–278 (2017).
26. Comeau, S. R., Gatchell, D. W., Vajda, S. & Camacho, C. J. ClusPro: a fully automated algorithm for protein–protein docking. *Nucleic Acids Res.* **32**, W96–W99 (2004).
27. Comeau, S. R., Gatchell, D. W., Vajda, S. & Camacho, C. J. ClusPro: an automated docking and discrimination method for the prediction of protein complexes. *Bioinformatics* **20**, 45–50 (2004).
28. Drake, Z. C., Seffernick, J. T. & Lindert, S. Protein complex prediction using Rosetta, AlphaFold, and mass spectrometry covalent labeling. *Nat. Commun.* **13**, 7846 (2022).
29. Russel, D. et al. Putting the pieces together: integrative modeling platform software for structure determination of macromolecular assemblies. *PLoS Biol.* **10**, e1001244 (2012).
30. Molnar, K. S. et al. Cys-scanning disulfide crosslinking and Bayesian modeling probe the transmembrane signaling mechanism of the histidine kinase, PhoQ. *Structure* **22**, 1239–1251 (2014).
31. Bayrhuber, M. et al. Structure of the human voltage-dependent anion channel. *Proc. Natl Acad. Sci. USA* **105**, 15370–15375 (2008).
32. Dauden, M. I. et al. Architecture of the yeast Elongator complex. *EMBO Rep.* **18**, 264–279 (2017).
33. Sirui, L. et al. Assisting and accelerating NMR assignment with restrained structure prediction. *Commun. Biol.* **8**, 1067 (2025).
34. Chaudhury, S., Lyskov, S. & Gray, J. J. PyRosetta: a script-based interface for implementing molecular modeling algorithms using Rosetta. *Bioinformatics* **26**, 689–691 (2010).
35. Alford, R. F. et al. Correction to ‘The Rosetta all-atom energy function for macromolecular modeling and design’. *J. Chem. Theory Comput.* **18**, 4594 (2022).
36. Lai, J. K., Ambia, J., Wang, Y. & Barth, P. Enhancing structure prediction and design of soluble and membrane proteins with explicit solvent–protein interactions. *Structure* **25**, 1758–1770.e1758 (2017).
37. Bartolec, T. K. et al. Cross-linking mass spectrometry discovers, evaluates, and corroborates structures and protein–protein interactions in the human cell. *Proc. Natl Acad. Sci. USA* **120**, e2219418120 (2023).
38. Wang, L., Xie, L. & Zhang, Z. Determination of HER2 binding domain in antigen–antibody complexes based on chemical crosslinking mass spectrometry. *J. Proteom.* **286**, 104954 (2023).
39. Di Ianni, A., Di Ianni, A., Cowan, K., Barbero, L. M. & Sirtori, F. R. Leveraging cross-linking mass spectrometry for modeling antibody–antigen complexes. *J. Proteome Res.* **23**, 1049–1061 (2024).
40. Wang, X. et al. Molecular details underlying dynamic structures and regulation of the human 26S proteasome. *Mol. Cell Proteom.* **16**, 840–854 (2017).
41. Kalisman, N., Adams, C. M. & Levitt, M. Subunit order of eukaryotic TRiC/CCT chaperonin by cross-linking, mass spectrometry, and combinatorial homology modeling. *Proc. Natl Acad. Sci. USA* **109**, 2884–2889 (2012).
42. Blech, M. et al. One target—two different binding modes: structural insights into gevokizumab and canakinumab interactions to interleukin-1 $\beta$ . *J. Mol. Biol.* **425**, 94–111 (2013).
43. Vreven, T. et al. Updates to the integrated protein–protein interaction benchmarks: Docking Benchmark Version 5 and Affinity Benchmark Version 2. *J. Mol. Biol.* **427**, 3031–3041 (2015).
44. Hitawala, F. N. & Gray, J. J. What has AlphaFold3 learned about antibody and nanobody docking, and what remains unsolved? Preprint at *bioRxiv* <https://doi.org/10.1101/2024.09.21.614257> (2024).
45. Cao, Y. et al. Imprinted SARS-CoV-2 humoral immunity induces convergent Omicron RBD evolution. *Nature* **614**, 521–529 (2023).
46. Ruffolo, J. A., Chu, L. S., Mahajan, S. P. & Gray, J. J. Fast, accurate antibody structure prediction from deep learning on massive set of natural antibodies. *Nat. Commun.* **14**, 2389 (2023).

47. Kosinski, J. et al. Molecular architecture of the inner ring scaffold of the human nuclear pore complex. *Science* **352**, 363–365 (2016).
48. Jamali, K. et al. Automated model building and protein identification in cryo-EM maps. *Nature* **628**, 450–457 (2024).
49. Kaake, R. M. et al. Characterization of an A3G-Vif<sub>HIV-1</sub>-CRL5-CBF $\beta$  structure using a cross-linking mass spectrometry pipeline for integrative modeling of host-pathogen complexes. *Mol. Cell. Proteomics* **20**, 100132 (2021).
50. Kuenze, G. et al. Allosteric mechanism for KCNE1 modulation of KCNQ1 potassium channel activation. *eLife* **9**, e57680 (2020).
51. Sun, J. & MacKinnon, R. Structural basis of human KCNQ1 modulation and gating. *Cell* **180**, 340–347 e349 (2020).
52. Escobar, C. A. et al. Structural interactions define assembly adapter function of a type II secretion system pseudopilin. *Structure* **29**, 1116–1127 e1118 (2021).
53. Banchenko, S. et al. Structural insights into Cullin4-RING ubiquitin ligase remodelling by Vpr from simian immunodeficiency viruses. *PLoS Pathog.* **17**, e1009775 (2021).
54. Wheat, A. et al. Protein interaction landscapes revealed by advanced in vivo cross-linking-mass spectrometry. *Proc. Natl Acad. Sci. USA* **118**, e2023360118 (2021).
55. Chen, Y. et al. Targeted cross-linker delivery for the in situ mapping of protein conformations and interactions in mitochondria. *Nat. Commun.* **14**, 3882 (2023).
56. Rath, S. et al. MitoCarta3.0: an updated mitochondrial proteome now with sub-organelle localization and pathway annotations. *Nucleic Acids Res.* **49**, D1541–D1547 (2021).
57. Merkley, E. D. et al. Distance restraints from crosslinking mass spectrometry: mining a molecular dynamics simulation database to evaluate lysine-lysine distances. *Protein Sci.* **23**, 747–759 (2014).
58. Yoshida, S. et al. Molecular chaperone TRAP1 regulates a metabolic switch between mitochondrial respiration and aerobic glycolysis. *Proc. Natl Acad. Sci. USA* **110**, E1604–E1612 (2013).
59. Thomas, P. D. The gene ontology and the meaning of biological function. *Methods Mol. Biol.* **1446**, 15–24 (2017).
60. Bulbarelli, A., Valentini, A., DeSilvestris, M., Cappellini, M. D. & Borgese, N. An erythroid-specific transcript generates the soluble form of NADH-cytochrome b5 reductase in humans. *Blood* **92**, 310–319 (1998).
61. Shor, B. & Schneidman-Duhovny, D. CombFold: predicting structures of large protein assemblies using a combinatorial assembly algorithm and AlphaFold2. *Nat. Methods* **21**, 477–487 (2024).

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2025

## Methods

### Datasets

**Training dataset.** We used the PSP dataset<sup>62</sup> as the basis for the GRASP and MEGAFold-Multimer training set. The PSP dataset contains a 570,000 true structure sets released before 8 October 2021 with protein sequences, experimental structures, MSAs and templates, along with a 760,000 distillation set with predicted structures, MSAs and templates. Complex samples were constructed by pairing single-chain MSAs with the same PDB ID in the true structure set using the pairing protocol within AFM<sup>4</sup> and the rest of the single-chain information combined.

**Self-curated test dataset.** To assess the ability of GRASP to integrate two types of restraint (contact RPR and IR), we curated a test dataset and benchmarked GRASP against several other methods.

**Sample collection.** To ensure a comprehensive and unbiased evaluation, we created a new challenging and high-quality dataset to avoid data leakage and redundancy. We retrieved the protein sequences for each PDB ID released before 20 March 2023, split them into individual chains, and excluded nonprotein chains and those shorter than 20 residues, resulting in about 707,901 single-chain protein sequences. Using ‘easy cluster’ in MMSeqs2 (ref. 63) with a 70% sequence identity threshold, we clustered these sequences into 58,671 clusters.

To avoid data leakage, we removed all protein complexes released before 8 October 2021, as well as complexes with the same cluster composition as those released before this date. We note here that we considered only the cluster type. Only biological assembly 1 was used. We then filtered complexes by length, resolution and chain information (Supplementary Methods). This process resulted in 1,336 protein complexes. We further refined this set to 313 complexes where the AFM prediction had a DockQ score below 0.23. We focus solely on interfaces in this part. For complexes with three or more chains, we randomly sampled up to five nonredundant pairs of contacting chains, resulting in a total of 713 interfaces.

**Contact RPR restraint sampling.** The 313 protein complexes were randomly divided into 3 groups, and each interface in the 3 groups was assigned with 2, 5 or 10 contact RPRs, respectively. Contact RPRs were sampled from inter-chain residue pairs with C $\beta$  distances within 8 Å, and no erroneous restraints were introduced.

**IR sampling.** Similarly, the 313 protein complexes were randomly divided into 3 groups, with interfaces in each group assigned with 4, 10 or 20 IRs. IRs were randomly sampled from interface residues whose C $\beta$  atoms are within 8 Å of the C $\beta$  atoms of another chain, without introducing erroneous restraints.

For both contact RPR and IR, we treat each chain as a distinct entity and apply restraints directly to the chains from which they were sampled, without distinguishing between homomeric and heteromeric chains.

**Simulated XL dataset.** To assess the performance of GRASP on simulated XL data, we generated XLs for the 313 complexes curated in the previous section. We performed inter-chain XL simulations using Xwalk<sup>64</sup>, applying a C $\beta$ –C $\beta$  cutoff of 25 Å and trypsin digestion. Three sets of restraints were simulated, with crosslinks (XLs) amounting to 1%, 2% and 5% of the total sequence length. We treat all chains as distinct entities and apply restraints directly to them, regardless of homomeric or heteromeric type. To mimic real experimental conditions, we included 20% erroneous restraints (false discovery rate of 20%), defined as residue pairs with C $\beta$ –C $\beta$  distances exceeding 25 Å, where at least one residue is Lys, Ser, Thr or Tyr. After excluding 19 complexes having fewer than 40% of the desired correct XLs, 294 complexes remained for analysis.

**Experimental datasets with single restraint type.** We constructed datasets with single experimental restraint types to compare the

performance of different methods. These include nine XL, five CL and four CSP cases.

**XL dataset.** The dataset consists of five antigen–antibody complexes<sup>38,39</sup> (PDB 1YY9, 3C09, 4G3Y, 6OGE\_ABC and 6OGE\_ADE) and four large protein complexes<sup>21,40,41</sup> (PDB 4B2T, 5L4K, 6WVK and 6ZX9). XLs were transformed into RPRs with distance range determined by the specific XL reagents, for example, 24 Å for disuccinimidyl sulfoxide (see Supplementary Table 7 for the XL distances for different reagents). For PDB 6OGE, We predicted HER2 extracellular domain binding to Pertuzumab and Trastuzumab, respectively, and named the two complexes PDB 6OGE\_ABC and 6OGE\_ADE following ref. 38. PDB 4B2T has a full length of 8,682 residues, which is too large for our model. Since it is of C2 symmetry, we chose to make structure prediction on the nonsymmetric half.

**CL dataset.** The five cases of CL were collected from ref. 28 (PDB 1YAG, 2F80, 4INS for 4INS4, 4INS8 and 4INS12), as we did in our ColabDock model<sup>22</sup>. Covalent-labeling data provide information on residues buried in the binding interface and can be naturally converted into IRs. The prediction results of ColabDock, HADDOCK and ClusPro are available from ref. 22.

**CSP dataset.** We obtained four CSP examples, two from ref. 24 (PDB 1GGR and 3EZA) and the other two from ref. 42 (PDB 4G6M and 4G6J). The CSP data are transferred into the form of IRs. Since PDB 4G6M and 4G6J are antibody–antigen systems and only have restraints on the antigen side, we selected the residues in the CDR region as IRs on the antibody side. The antibody CDR regions were identified using the ‘Chothia’ scheme ([www.bioinf.org.uk/abs/abnum/](http://www.bioinf.org.uk/abs/abnum/))<sup>65</sup>. The prediction results of 1GGR and 3EZA of ColabDock, HADDOCK and ClusPro were acquired from ref. 22. For PDB 4G6M and 4G6J, we ran calculations using these tools with the same settings as used in ref. 22 to complete the benchmark.

**Simulated DMS dataset. BMS.5.** To evaluate GRASP on antibody–antigen systems, we simulated three sets of DMS restraints for each of the 67 antibody–antigen complexes in the BMS.5 (ref. 43) dataset. Of these, 45 complexes had already been assessed by ref. 22, and we reused the simulated restraints for them. The remaining 22 complexes were processed using the same restraint simulation strategy as in ref. 22.

**Hitawala–Gray dataset.** To prevent data leakage, we simulated DMS restraints using the case list curated in ref. 44, which includes 118 antibodies and 70 nanobodies released after the 2,021 AF3 training cutoff. Two antibodies (PDB 7YIZ and 8CBY) were excluded due to the absence of antigen chains, and 9 antibodies and 5 nanobodies were removed due to duplication, resulting in 107 antibodies and 65 nanobodies. A random selection of 5–10 interface sites was sampled on each side for the simulation. We also evaluated AF3’s performance on this dataset.

**Experimental DMS dataset.** We obtained DMS profiles for 3,051 SARS-CoV-2 wild-type RBD-targeting monoclonal antibodies from ref. 45. We then searched the PDB<sup>3</sup> database using the monoclonal antibody names to identify protein complexes that met our criteria, specifically those containing the spike glycoprotein, light chain and heavy chain. This search resulted in a final set of 84 complexes (Supplementary Data). The spike glycoproteins were aligned to the Wuhan-Hu-1 RBD sequence (residues N331 to T531, GenBank MN908947), and mutation escape sites for each complex were determined. We trimmed the spike protein to include only the RBD regions for further analysis. The antibody CDR regions were identified using the Chothia scheme ([www.bioinf.org.uk/abs/abnum/](http://www.bioinf.org.uk/abs/abnum/))<sup>65</sup>.

**Mitochondria dataset.** We collected targeted XL-MS data from ref. 55 and kept only the inter-protein XLs. We merged XLs with the same

gene name to improve the restraint quantity for each pair of PPIs. We renumbered the crosslinked sites according to corresponding sequence in human MitoCarta v.3.0 database<sup>56</sup> by sequence alignment and prepared the sequences for structure prediction (Supplementary Data). This led to a dataset including 144 PPIs with 1–4 pairs of RPRs for each. We set the maximum C $\beta$ –C $\beta$  distance to 30 Å for disuccinimidyl suberate to be consistent with ref. 55.

### Model training strategy

**MEGAFold-Multimer training.** We trained a MEGAFold-Multimer model from scratch using 128 Ascend-910A (32G) AI chips. MEGAFold-Multimer is a reimplementation of AFM within the MindSpore framework. The training process consisted of two stages (46,200 total steps over 17 days): an initial training phase (33,000 steps over 12 days) and a fine-tuning phase (13,200 steps over 5 days). Hyperparameters and training details are provided in the Supplementary Methods.

**GRASP model fine-tuning.** We fine-tuned on both AFM v.2.3 Model-1 checkpoint and MEGAFold-Multimer checkpoint over 9 days separately using 64 Ascend-910A (32G) AI chips. To achieve a more stable prediction, we generated five models. Four of these were fine-tuned from AFM v.2.3 Model 1, taken at 8,000 (JAX-8k), 14,000 (JAX-14k), 20,000 (JAX-20k) and 22,000 (JAX-22k) steps during the fine-tuning process. The fifth model (MS-22k) was fine-tuned 22,000 steps from the MEGAFold-Multimer model.

We adjusted the multimer/monomer ratio to 4/1 in GRASP training, set the learning rate to  $1 \times 10^{-4}$ , and used a crop size of 384. All other configurations were consistent with those of AFM unless specified otherwise. We also introduced four restraint-related loss terms during fine-tuning, including RPR FAPE loss, RPR distance r.m.s.d. loss, RPR distogram loss and the interface loss (details in Supplementary Methods).

We simulated intra-chain RPRs (intra-RPRs), inter-chain RPRs (inter-RPRs) and IRs during fine-tuning, separately. Intra-RPR and inter-RPR were handled separately considering their differences in distance distribution, quantity and impact on the predicted structure. The details for sampling number, order and form of each type of restraints are provided in Supplementary Methods and Supplementary Fig. 21.

**Iterative restraint filtering.** To address noise in experimental restraint signals in model inference, which can lead to inaccurate structure predictions, we implement an iterative restraint filtering strategy (Supplementary Fig. 22a). Each round of restraint filtering evaluates the validity of the current restraints based on the predicted structure. Two key metrics are assessed in the order of priority: (1) chain break at the restrained residues and (2) the extent to which the structure violates the restraint.

We regard the C $\alpha$  distance between consecutive residues higher than 5 Å as a chain break. The neighboring C $\alpha$  distance ( $d_{\text{NBCA}}$ ) for a residue is defined as the average C $\alpha$  distance to its two (one for terminal residues) flanking residues. The  $d_{\text{NBCA}}$  for a RPR restraint is defined as the average  $d_{\text{NBCA}}$  of the two restrained residues, while for an IR restraint it is the  $d_{\text{NBCA}}$  of the interface residue.

The violation distance ( $d_{\text{viol}}$ ) measures how much the predicted structure violates the restraint. For example, if an IR specifies residue  $i$  as an interface residue but its closest distance to another chain is 20 Å,  $d_{\text{viol}}$  is 12 Å. For an RPR restraint requiring residues  $i$  and  $j$  to be within 25 Å,  $d_{\text{viol}}$  is 0 if their predicted distance is 10 Å, and 5 Å if the distance is 30 Å.

Using these two distances, the irrationality score (IS) of a restraint is defined as:

$$\text{IS} = \max(d_{\text{NBCA}} - 5, 0) \times 1000 + \max(d_{\text{viol}} - 5, 0) \quad (1)$$

To avoid excessive restraint removal, each iteration filters up to 20% of current restraints, ensuring at least 20% of the original restraints

are retained. The filtering stops when the maximum iterations are reached, no restraints are removed or fewer than 20% of the original restraints remain. IR and RPR filtering are executed independently, and the final predicted structure is then output.

Convergence is typically reached within 2–3 iterations, but a larger number may be needed with a large number of restraints. To improve efficiency, we introduced a ‘fast inference mode’ (Supplementary Fig. 22b). This mode reduces the number of recycles between iterations and recycles the structure after each filtering step, allowing up to 20 inference rounds and making it up to 4 times faster than the standard version. We used this mode in the DMS dataset evaluation, where RPRs across both sides of the interface introduced  $m \times n$  RPRs, with  $m$  and  $n$  representing the number of interfaces on each side.

**Integrative modeling using Combfit.** To smoothly integrate cryo-EM density map information into GRASP, we developed Combfit, an advancement over the GPU-based matching software Powerfit<sup>66</sup>, using clustering and genetic algorithms (Supplementary Methods). Implemented in up-to-date Python v.3.10, Combfit offers excellent readability and is easy to integrate with other models.

In integrative modeling involving cryo-EM, we first converted all restraints, except for the density map, into either RPRs (for XL and NMR nuclear Overhauser effect spectroscopy) or IRs (for CL, mutagenesis and CSP) to predict the initial structure. The initial structure was then split into random or predefined chain combinations, which we refer to as ‘random-split’ strategy. These chain combinations were docked into density map using Combfit. We next extracted residue pairs from the docked structure where the C $\beta$ –C $\beta$  distance was  $\leq 25$  Å and had decreased by at least 10 Å compared to the initial structure. Using a min–max algorithm<sup>67</sup>, up to 30 representative residue pairs were chosen from the extracted residue pairs as RPRs. These RPRs, along with the existing RPRs and IRs from other experiments, were then used to guide the GRASP’s second round of prediction. In doing so, we obtained the final structure for integrative modeling with cryo-EM data.

**Setup for GRASP inference.** The AFM protocol<sup>4</sup> was used to search for MSA and templates and to prepare input features for GRASP. The maximum template date was set to ‘2021-10-08’. We used reduced databases in cases where HHblits raised errors during MSA searches. The standard iterative restraint filtering strategy was applied with a maximum of 5 iterations for cases with fewer than 3,072 residues and 2 iterations for larger ones. Each checkpoint runs with 5 seeds for complexes under 3,072 residues to align with the AFM setup and 1 seed for larger complexes. For all the predicted structures, we select the predicted structure with the highest pLDDT score as the final prediction, prioritizing those with a recall of at least 30%. Due to memory limitations, the Ascend-910A (32G) could not handle inference for sequences longer than 1,280 residues. To address this, we migrated GRASP’s code and checkpoint to JAX (GRASP-JAX). For complexes exceeding 1,280 in length, we performed inference using GRASP-JAX on A100 (80G) GPUs. To reduce computational costs, structure relaxation was disabled by default, except when evaluating interface-specific physicochemical features (for example, hydrogen bonding), in which case the default relaxation protocol was applied to add hydrogen atoms. The basic setup described above was used for all GRASP inference, with some unique settings as follows:

For the experimental DMS dataset, we used both RPR and IR (Fig. 4) and applied quick mode iterative filtering to speed up inference (‘Iterative restraint filtering’ section). We also evaluated performance when IR and RPR were used separately (Supplementary Fig. 23).

For the simulated DMS dataset, we divided all complexes into antigen and antibody for docking. We used GRASP in Dimer Mode, which converts complexes with more than two chains into dimers by setting the same ‘asym\_id’ for chains in each group and adding a 200-residue gap between chains<sup>61</sup>. This approach improves IR precision, narrows

the search space, and simplifies the task. GRASP's performance without Dimer Mode was also evaluated (Supplementary Fig. 9). For fair comparison, all other methods were also evaluated by splitting complexes into antigen and antibody.

For the integrative modeling cases, we applied the integrative strategy with Combfit mentioned above to incorporate additional information from density map for A3G-VCBC. For the rest two cases standard setup of GRASP was used. For the scoring stage of all three cases, we normalized each structure's pLDDT and recall by subtracting the mean and dividing by the standard deviation, then selected the one with the highest average normalized score.

For the inference of the dimeric interactions in the mitochondria, we disabled the iterative restraint filtering strategy due to the sparsity of the restraints, the number of which is only 1–4 per complex.

**Configurations for other structure prediction tools.** *AFM.* We used the same input features for AFM as GRASP. Five v.2.3 models were used, with each model predicting 5 structures for sequences up to 3,072 residues and 1 structure for longer sequences due to limitation in resource.

*AlphaLink.* We used the same input features for AlphaLink as GRASP. Inference was performed using the AlphaLink-Multimer\_SDA\_v2.pt checkpoint selected based on a prebenchmark on the 5% coverage simulated Sulfo-SDA restraints (Supplementary Fig. 24). To ensure a fair comparison, AlphaLink was run with an adjusted number of random seeds to match the number of generated structures in GRASP: 25 random seeds for complexes with up to 3,072 residues and 5 seeds for larger complexes.

*ColabDock.* The settings of ColabDock is the same as in ref. 22. We use AF2 (ref. 68) to generate single-chain templates and AFM to generate templates for the antibody side. If only one side of the interface was available, the IRs for the other side were generated by selecting residues with solvent accessibility  $\geq 40\%$  using the FreeSASA<sup>69</sup> software. These single-chain templates and processed restraints also serve as inputs of HADDOCK and ClusPro. Please note that since GRASP does not necessarily require single-chain templates and readily handles one-sided interfaces, these two types of information are not needed. For complexes whose total length exceeds 700 residues, we ran segment-based optimization with recommended settings.

*HADDOCK.* We installed a local version HADDOCK v.2.4 to perform benchmark experiments. We followed the default setting of HADDOCK and picked the best structure by HADDOCK score from 'water' refinement stage as the final structure. For cases that failed in the 'itl' or 'water' refinement stage due to energy optimization issues, we picked the best structure in the last stage. All RPRs were set as unambiguous restraints, while IRs were set as ambiguous restraints.

*ClusPro.* We installed the program PIPER provided on the ClusPro website and convert the restraints into json format using official scripts. We set the restraints satisfaction ratio to 50% for IRs and decreased this ratio to 30% for cases without available output structure at 50% ratio.

*AF3.* We used the web server (<https://golgi.sandbox.google.com/>) for our predictions on experimental XL, CL and DMS datasets. After the code was open-sourced, we also benchmarked AF3 on the self-curated test set and the simulated DMS dataset (Hitawala–Gray dataset) using the released code (<https://github.com/google-deepmind/alphafold3>). Five random seeds generated 25 predictions, with the highest-ranking result selected. All other settings were default.

*IMP.* We installed IMP v.2.22.0 in local machine. Single templates are predicted by AlphaFold. We treated each chain in a protein complex as a rigid body and transformed the initial density map

to three-dimensional (3D) gaussians to guide the modeling. The hyperparameters follow the tutorial of RNA Pol II on the IMP website unless otherwise stated.

## Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

All relevant evaluation datasets and results are available via OSF at <https://doi.org/10.17605/OSF.IO/6KJUQ> (ref. 70). Further training data for MEGAFold-Multimer and GRASP are released at <http://ftp.cbi.pku.edu.cn/pub/psp/>.

## Code availability

The GRASP code is available via GitHub at <https://github.com/xiergo/GRASP-JAX> and via Zenodo at <https://doi.org/10.5281/zenodo.15347070> (ref. 71) under the Apache v.2.0 license. Model weights have been deposited on OSF at <https://doi.org/10.17605/OSF.IO/6KJUQ> (ref. 70). In addition, a Colab notebook is available via GitHub at <https://colab.research.google.com/github/xiergo/GRASP-JAX/blob/main/GRASP-JAX.ipynb> for ease of use.

## References

62. Liu, S. et al. PSP: million-level protein sequence dataset for protein structure prediction. Preprint at <https://arxiv.org/abs/2206.12240> (2022).
63. Steinegger, M. & Söding, J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat. Biotechnol.* **35**, 1026–1028 (2017).
64. Kahraman, A., Malmström, L. & Aebersold, R. Xwalk: computing and visualizing distances in cross-linking experiments. *Bioinformatics* **27**, 2163–2164 (2011).
65. Abhinandan, K. R. & Martin, A. C. Analysis and improvements to Kabat and structurally correct numbering of antibody variable domains. *Mol. Immunol.* **45**, 3832–3839 (2008).
66. Gydo, C. P. Z. & Alexandre, M. J. J. B. Fast and sensitive rigid-body fitting into cryo-EM density maps with PowerFit. *AIMS Biophys.* **2**, 73–87 (2015).
67. Eldar, Y., Lindenbaum, M., Porat, M. & Zeevi, Y. Y. The farthest point strategy for progressive image sampling. *IEEE Trans. Image Process.* **6**, 1305–1315 (1997).
68. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
69. Mitternacht, S. FreeSASA: an open source C library for solvent accessible surface area calculations. *F1000Res.* **5**, 189 (2016).
70. Xie, Y. et al. Datasets and model weights for GRASP. OSF <https://doi.org/10.17605/OSF.IO/6KJUQ> (2025).
71. Xie, Y. et al. Source code for GRASP. Zenodo <https://doi.org/10.5281/zenodo.15347070> (2025).

## Acknowledgements

This work was supported by the National Science and Technology Major Project (grant no. 2022ZD0115001 to Y.Q.G. and S. Liu), the National Natural Science Foundation of China (grant nos. 92353304 and T2495221 to Y.Q.G.) and New Cornerstone Science Foundation (grant no. NCI202305 to Y.Q.G.). We thank K. Stahl from Technische Universität Berlin for recommending AlphaLink configuration and providing simulated CASP15 restraints for debugging. We thank F. N. Hitawala from Johns Hopkins University for sharing the PDB IDs and sequence data of their curated benchmarking dataset. We thank Y. Kaiguang from Dalian Institute of Chemical Physics to provide guidance for usage of the targeted XL-MS in the mitochondria. We also thank Z. Zhu for helpful discussions on binding energy calculation.

## Author contributions

Y.Q.G. and S. Liu designed and developed overall concepts in the paper and supervised the project. Y.X., C.Z. and X.D. developed the GRASP and Combfit methods. M.W. and Y.H. trained and evaluated MEGAFold-Multimer. Y.X., C.Z., S. Li, X.D., Y.L. and Z.C. performed data collection, evaluation and analysis. X.Y., C.Z., S. Li and X.D. wrote the initial draft of the paper. All authors contributed ideas to the work and assisted in editing and revision.

## Competing interests

Changping Laboratory and Huawei Technologies Co., Ltd are in the process of applying for a patent covering the GRASP method, which lists authors including S. Liu, Y.X., C.Z. and Y.Q.G. as inventors. The other authors declare no competing interests.

## Additional information

**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1038/s41592-025-02820-1>.

**Correspondence and requests for materials** should be addressed to Sirui Liu or Yi Qin Gao.

**Peer review information** *Nature Methods* thanks the anonymous reviewers for their contribution to the peer review of this work. Primary Handling Editor: Arunima Singh, in collaboration with the *Nature Methods* team.

**Reprints and permissions information** is available at [www.nature.com/reprints](http://www.nature.com/reprints).

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

|                                     |                                                                                                                                                                                                                                                                                                |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n/a                                 | Confirmed                                                                                                                                                                                                                                                                                      |
| <input type="checkbox"/>            | <input checked="" type="checkbox"/> The exact sample size ( <i>n</i> ) for each experimental group/condition, given as a discrete number and unit of measurement                                                                                                                               |
| <input type="checkbox"/>            | <input checked="" type="checkbox"/> A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly                                                                                                                                    |
| <input type="checkbox"/>            | <input checked="" type="checkbox"/> The statistical test(s) used AND whether they are one- or two-sided<br><i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>                                                               |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> A description of all covariates tested                                                                                                                                                                                                                                |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons                                                                                                                                                   |
| <input type="checkbox"/>            | <input checked="" type="checkbox"/> A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| <input type="checkbox"/>            | <input checked="" type="checkbox"/> For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i> ) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted<br><i>Give P values as exact values whenever suitable.</i>                     |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings                                                                                                                                                                      |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes                                                                                                                                                |
| <input type="checkbox"/>            | <input checked="" type="checkbox"/> Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i> ), indicating how they were calculated                                                                                                                                               |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data collection | <p>For the self-curated test dataset, MMseqs2 (commit 6b9388469a41935813f5d207d6d149767feb4c42) was used for sequence clustering during sample collection. For the simulated XL dataset, Xwalk (v0.6) was employed to simulate XL restraints. Biopython (v1.81) was used for PDB data cleaning and processing, and NumPy (v1.22.4) was used for data preprocessing and restraint sampling.</p> <p>For the experimental datasets, the web server <a href="http://www.bioinf.org.uk/abs/abnum/">http://www.bioinf.org.uk/abs/abnum/</a> was used to identify CDR regions in antibodies. FreeSASA (v2.1.2) generated IRs on the opposite side for 1YAG with one-sided IRs for ColabDock, HADDOCK, and ClusPro. ChimeraX (v1.9) was used to generate lower-resolution density maps for A3G-VCBC, and EMAN2 (v2.99.47) was used to measure the resolution of the density maps. The Python interface for the RCSB PDB search API, "rcsbsearchapi" (v1.6.0), was used to retrieve experimentally resolved structures from the PDB dataset for the integrative modeling and mitochondria datasets.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Data analysis   | <p>The GRASP code is available at <a href="https://github.com/xiergo/GRASP-JAX">https://github.com/xiergo/GRASP-JAX</a>; see the Code Availability section for details. HADDOCK (v2.4) and ClusPro (v2.0) were locally installed and configured following official guidelines. For AlphaFold and AlphaFold-Multimer, code from <a href="https://github.com/google-deepmind/alphafold">https://github.com/google-deepmind/alphafold</a> (v2.3.3) was used with default settings to generate single-chain and complex predictions. ColabDock (v2.0) was run with reference configurations. AlphaLink (<a href="https://github.com/Rappsilber-Laboratory/AlphaLink2">https://github.com/Rappsilber-Laboratory/AlphaLink2</a>, v1.1.1) was used with the AlphaLink-Multimer_SDA_v2.pt checkpoint, adjusting random seeds to align with GRASP's structure generation approach. AlphaFold3 (AF3) was run on the web server (<a href="https://golgi.sandbox.google.com/">https://golgi.sandbox.google.com/</a>) for the XL/CL/DMS datasets using default settings. After open-sourcing, AF3 was benchmarked on the self-curated test set and the simulated DMS dataset (Hitawala-Gray) using the released code (<a href="https://github.com/google-deepmind/alphafold3">https://github.com/google-deepmind/alphafold3</a>, v3.0.0), with five random seeds generating 25 predictions and the best result selected. IMP (v2.22.0) was used for integrative modeling benchmarks. PowerFit (v2.1.0) was converted to Python3 and extended into Combfit, which was used for cryo-EM data integration. Detailed methodological setups are described in the Methods and Supplementary Methods.</p> <p>Structure evaluation was primarily based on DockQ scores (<a href="https://github.com/bjornwallner/DockQ">https://github.com/bjornwallner/DockQ</a>, v1.0), TM-Score (<a href="https://zhanggroup.org/">https://zhanggroup.org/</a>)</p> |

MM-align/, version 20210816), IDDT (v1.3), RMSD, and restraint recall, with the latter two computed using custom scripts. Additional analyses, such as calculating averages and medians, were performed using Python tools. Physicochemical properties were computed using PyRosetta (v2024.19+release.a34b73c40f). Statistical analyses, including Pearson and Spearman correlation coefficients and one-sided pairwise t-tests, were conducted using the SciPy package (v1.6.3). PyMOL (v2.5.5) was used for 3D structure visualization. Matplotlib (v3.5.1) and Seaborn (v0.13.0) were used for data visualization.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

## Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All related test datasets are available at <https://doi.org/10.17605/OSF.IO/6KJUQ>. The Hitawala–Gray dataset, an unpublished dataset curated by Hitawala and Gray, will be released once the original dataset is officially published. Additional training data for MEGA-Fold-Multimer and GRASP are available at <http://ftp.cbi.pku.edu.cn/pub/psp/>.

## Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

N/A

Reporting on race, ethnicity, or other socially relevant groupings

N/A

Population characteristics

N/A

Recruitment

N/A

Ethics oversight

N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

## Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

## Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

The PSP training set was compiled from previous work and provides ample training data. The Self-Curated Test Dataset was constructed by us, with careful removal of redundancy against both the training set and within itself. After applying quality control on sequence length, chain number, and resolution, we retained all remaining challenging samples (AFM DockQ < 0.23), yielding 713 interfaces from 313 complexes. Following further filtering based on the number of cross-links, 294 complex samples were selected, on which three sets of Sulfo-SDA simulated restraints (1%, 2%, and 5% coverage) were applied. This dataset is sufficient to evaluate the performance of various tools under different restraint levels.

The experimental XL dataset consists of five antigen–antibody complexes and four protein complexes (nine in total). The CL dataset includes five covalent labeling cases from published studies, and the CSP dataset contains four chemical shift perturbation cases from the literature. The simulated DMS dataset includes two components: (1) the BM5.5 dataset with 67 antibody–antigen complexes from the public benchmark 5.5 dataset, each with three simulated restraint sets; and (2) the Hitawala–Gray dataset with 107 antibodies and 65 nanobodies, each with one simulated restraint set. The experimental DMS dataset includes 84 complexes with deep mutational scanning profiles for SARS-CoV-2 RBD-targeting monoclonal antibodies from published studies. The mitochondria dataset comprises 144 protein–protein interactions (PPIs), each with one to four restraints, derived from human mitochondrial cross-linking data by Chen et al. The integrated modeling dataset includes four protein complexes with restraints from multiple experimental sources, including chemical cross-linking, mutagenesis, electron density maps, custom templates, and NMR paramagnetic relaxation enhancement (PRE).

These experimental datasets and the original cases for simulated DMS dataset were collected from prior studies, and their sample sizes were determined by the scope of the original works. The filtering criteria are detailed in the “Data Exclusion” section.

Overall, these datasets provide broad coverage of protein complex types, restraint conditions, and biological contexts. Together, they offer

sufficient diversity and scale to robustly evaluate the generalizability, accuracy, and versatility of the proposed approach across both simulated and experimental settings.

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data exclusions | <p>For the construction of the Self-Curated Test Dataset, we removed all protein complexes released before October 8, 2021, and excluded those sharing the same cluster composition as any entries from before this date to prevent data leakage. We then applied the following criteria to retain complexes in the self-curated dataset: 1) total chain length <math>\leq 5120</math> residues; 2) resolution better than 3.0 Å; 3) 2 to 20 chains; 4) shortest chain <math>\geq 32</math> residues; 5) every chain in contact with at least one other chain from the complex; 6) for duplicates with the same cluster composition, the complex with the smallest PDB ID was kept. The dataset was further refined by removing complexes with AFM DockQ scores greater than 0.23 to retain only the challenging cases. For multi-chain complexes, we limited the selection to up to five non-redundant pairs of contacting chains, resulting in a total of 713 interfaces sampled from 313 complexes.</p> <p>In the Simulated XL Dataset, 19 complexes were excluded for having fewer than 40% of the expected sampling number of crosslinks, resulting in selection of 294 complexes.</p> <p>For the Simulated DMS Dataset: 1). In the BM5.5 dataset, we removed complexes that were not antibody-antigen pairs. 2). In the Hitawala-Gray dataset, two antibodies (7YIZ and 8CBY) were excluded due to the absence of antigen chains, and 9 antibodies and 5 nanobodies were removed due to duplication.</p> <p>For the Experimental DMS Dataset, we excluded mAbs that could not be found in the PDB database.</p> <p>In addition, we exclude the cases each method fails in their corresponding statistics, and the excluded case number and id can be found in the manuscript, figure description, and extended data.</p> |
| Replication     | Based on the code in the github repository, when providing the same restraints and input sequence, all results could be reproduced by repetitive runs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Randomization   | <p>For the benchmark of various methods on contact-RPR and IR restraints, to evaluate model performance at different restraint numbers, we divided the samples into three groups, each assigned a different number of restraints. For Contact-RPR restraint sampling, the 313 protein complexes were randomly split into three groups, with each interface assigned either 2, 5, or 10 contact-RPRs. Similarly, for IR sampling, the complexes were randomly divided into three groups, with interfaces assigned 4, 10, or 20 IRs.</p> <p>For the other datasets, the same samples and restraints were used for comparison and evaluation of all applicable methods, with no randomization applied to the model input.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Blinding        | In method comparison, We wish to note that each sample in this study could be run independently and repetitively, and the same input data (for example, sequence, msa, template and restraint) can be seen by different methods. It is therefore impossible and unnecessary for blinding in comparison.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

### Materials & experimental systems

|                                     |                                                        |
|-------------------------------------|--------------------------------------------------------|
| n/a                                 | Involved in the study                                  |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Antibodies                    |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Eukaryotic cell lines         |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Palaeontology and archaeology |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Animals and other organisms   |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Clinical data                 |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Dual use research of concern  |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Plants                        |

### Methods

|                                     |                                                 |
|-------------------------------------|-------------------------------------------------|
| n/a                                 | Involved in the study                           |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> ChIP-seq               |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Flow cytometry         |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> MRI-based neuroimaging |

## Plants

|                       |     |
|-----------------------|-----|
| Seed stocks           | N/A |
| Novel plant genotypes | N/A |
| Authentication        | N/A |